

Announced Rescue Pricing: Strategic Waiting, Screening History, and Market Thickness

(Authors' names blinded for peer review)

A platform that announces a higher payment for an unfilled request improves its chance of rescue but also gives current suppliers a reason to wait. We study this trade-off when the expected thickness of the incumbent driver pool is public, its realization is latent, and universal rejection both triggers continuation and screens the remaining supply. A private-value rider decides whether to post and, after failure, whether to abandon, repeat, or activate rescue; private-cost incumbents decide whether to accept the initial payment or preserve a discounted option to wait. Within an anonymous symmetric cutoff-WPBE class, we characterize equilibrium with incumbent survival and fresh driver entry. In the uniform no-entry core, every menu has a unique cutoff and the two-payment design reduces to a one-dimensional optimization. The optimal rescue menu strictly improves completion above an endogenous patience threshold. Its completion gain relative to the optimal flat payment is continuous in market thickness, vanishes in both very thin and very thick markets, and therefore attains a positive maximum at a finite thickness; uniqueness of that maximizer is not asserted. With both surviving incumbents and fresh entrants, a small rescue strictly beats optimized flat pricing whenever at least one optimal flat price leaves the highest rider type strictly willing to use rescue. Holding either continuation headcount or price-marginal capacity fixed, a larger incumbent share strictly intensifies first-order strategic waiting within the active local-rescue region, while its completion effect can reverse with incumbent patience. We also provide a pre-specified model-to-data mapping for a request-level structural calibration; quantitative estimates are deliberately withheld until the announcement, exposure, and entry measurements required by the model are available.

Key words: dynamic pricing, rescue payments, strategic waiting, market thickness, screening, Poisson games

1. Introduction

Recent reporting makes the operational problem concrete. In January 2026, DoorDash warned that severe-weather demand could outstrip available couriers, producing long waits and cancelled orders; its stated response included a temporary fee passed through to active couriers as Peak Pay (DoorDash 2026). Reuters documented a sharper rejection mechanism in Kenya: some ride-hailing drivers asked riders to pay more than the app price and cancelled the request when the rider declined, sending the rider back to search for another driver (Okoth 2024). Continuation supply can change as well. Reuters reported that JD.com doubled its three-month recruitment target for full-time delivery riders from 50,000 to 100,000 (Reuters 2025). These episodes establish the separate forces of scarcity, rejection, higher compensation, rematching, and fresh entry; they do

not establish the exact same-request contract studied here. We ask whether committing ex ante to a higher payment after universal rejection rescues a request or instead teaches incumbents to wait for the next offer.

Many service requests are initially unattractive to available suppliers but become urgent after a failed response window. A platform can address this problem by announcing an initial payment and a higher, failure-contingent rescue payment. Announcement makes the policy credible, but it also changes the first response: a supplier may reject today in order to preserve the option of receiving more tomorrow. The platform must therefore design the rescue schedule and the induced waiting equilibrium jointly.

This tension is especially sharp when the platform observes an expected market state but not the realized local supply attached to a particular request. Universal rejection is then both an operational event and an informational event. It triggers the second response window and reveals that every incumbent who was willing to accept the initial payment is absent from the continuation pool. Incumbents who remain are selected by their earlier rejection, whereas drivers who enter later have not been screened. The value of future supply consequently depends on its screening history, not only on its expected quantity.

We study a two-period request in which a platform commits to payments (p_1, p_2) with $p_2 \geq p_1$. A private-value rider first decides whether to post. Private-cost incumbent drivers then simultaneously accept p_1 or wait. After universal rejection, the rider can abandon, repeat p_1 , or activate p_2 . Incumbents may survive to the second window and a fresh pool may enter. The expected incumbent thickness is public, while realized local supply and rival costs remain latent. Drivers who wait discount their terminal payoff, and accepting drivers compete for assignment.

The analysis follows a policy–response–optimization sequence. We first fix the announced menu and characterize weak perfect Bayesian equilibrium within the anonymous symmetric pure-cutoff class. We then optimize the menu subject to that equilibrium response. Universal rejection generates a tractable failure posterior: incumbents remaining above the first-period cutoff are a screened population, while fresh entrants retain the original cost distribution. The equilibrium characterization allows arbitrary incumbent survival and fresh entry and makes the rider’s abandon–repeat–rescue choice part of the drivers’ continuation payoff.

The globally solvable core sets fresh entry to zero while retaining incumbent survival. In the uniform model, every announced menu has a unique equilibrium cutoff. For a fixed p_1 , the optimal rescue problem separates into reject-all, tangent-rescue, and repeat-only regimes. This reduces joint design of p_1 and p_2 to a continuous one-dimensional problem and proves policy attainment. Above an endogenous rider-patience threshold, the optimized rescue menu strictly improves completion

relative to the optimized flat payment. Positive incumbent survival makes $\beta \geq 1/2$ a simple sufficient condition at every finite thickness.

We next return to the general continuation pool. When $\alpha < 1$ and $\gamma > 0$, the second window contains both screened surviving incumbents and unscreened new entrants. A local theorem shows when a small announced rescue increase improves completion after accounting for both sources of supply. If at least one optimized flat price satisfies this activity condition, the same perturbation proves strict policy-class dominance over optimized flat pricing, without requiring the flat optimizer to be unique. A separate composition theorem holds either the expected continuation headcount or the local price-marginal capacity fixed. Replacing entrants with screened incumbents strictly strengthens first-period acceptance erosion; its effect on the local completion gain has no unconditional sign and can reverse with δ .

Market thickness determines where rescue design matters. Let $M_R^*(m)$ and $M_F^*(m)$ denote the optimized completion probabilities under rescue and flat pricing at the same thickness. Their difference $V(m)$ is continuous, vanishes in very thin and very thick markets, and is strictly positive at some finite thickness. Hence a finite intermediate market maximizes the rescue gain. This existence result does not imply that $V(m)$ is strictly single-peaked or that the maximizing thickness is unique.

The paper also specifies a request-level empirical implementation. The data must record the announced menu, the initial exposure set, universal rejection, rider continuation, incumbent survival, fresh entry, and terminal completion. Those inputs are not present in the current repository. We therefore provide a pre-specified measurement map, calibration criterion, validation design, and counterfactual pipeline, with every empirical result explicitly marked DATA REQUIRED. Assumed-parameter computations remain theoretical numerical illustrations rather than empirical estimates.

This paper makes three contributions. First, it characterizes the cutoff-WPBE created by an announced same-request rescue menu, latent local competition, and a private rider continuation decision. Second, it solves the global no-entry design problem and establishes how patience and market thickness govern the completion gain from rescue pricing. Third, it separates screened surviving incumbents from unscreened fresh entrants, establishes conditional optimized-flat dominance in the mixed-supply model, and proves that composition matters even when continuation headcount or price-marginal capacity is held fixed. The contribution is the equilibrium and completion-design analysis generated by this joint structure, not the procurement-clock architecture, strategic waiting, entry, or market thickness in isolation.

The remainder of the paper reviews the closest literatures, states the empirical measurement requirements, develops the model and equilibrium, solves the design problem, studies thickness and supply composition, and lays out the calibration and policy-counterfactual procedure. Proofs and numerical verification details appear in the appendices.

2. Literature Review

Procurement clocks and strategic timing. Li and Kuo (2013) optimize a finite discrete Dutch auction that advances after universal nonacceptance, permits a Poisson number of bidders, and randomly selects among simultaneous acceptors. Reversing the transfer direction gives the low-to-high procurement skeleton of an announced rescue menu. Buchanan et al. (2016) derive symmetric Bayesian bidding when realized group size is hidden and nonacceptance changes beliefs along a multi-unit Dutch clock. Carare and Rothkopf (2005) study slow Dutch auctions with waiting costs, and Shneyerov (2014) optimizes a dynamic clock when market sides discount at different rates. These papers establish the importance of strategic timing, population uncertainty, and learning from nonacceptance. Our posted-payment environment instead embeds the clock in a private rider's continuation problem and evaluates unconditional request completion.

Search mechanisms provide a second foundation for the continuation pool. McAfee and McMillan (1988) study dynamic mechanisms that acquire additional participants over time. Crémer et al. (2007) formulate optimal search auctions in which previously contacted bidders are incumbents and later search brings new entrants; a rejected incumbent can therefore remain in a subsequent allocation with a newly solicited bidder. These papers establish retained incumbency and fresh entry before strategic delay is added.

Lee and Li (2023) is especially close to the survival-and-entry extension. Their seller commits to a reserve-price and sampling schedule; long-lived incumbent bidders can bid or wait, new bidders enter later, and earlier screening truncates the values of the incumbent pool while new bidders retain the original distribution. Under the procurement dual, this is the same screened-incumbent versus unscreened-entrant distinction that appears here. Their analysis establishes announced price paths, strategic waiting, and the mixed composition of future supply in a search-auction environment. Our setting instead uses an anonymous posted-payment allocation with latent Poisson competition, a rider who chooses among abandon, repeat, and rescue, and completion comparisons between optimized rescue and flat policies.

Announced dynamic pricing and forward-looking agents. Aviv and Pazgal (2008) solve announced markdown design by fixing a price path, deriving a forward-looking consumer threshold, and then optimizing the path. Correa et al. (2016) study contingent pricing with strategic consumers. These papers supply the methodological policy–response–optimization order, but their consumers wait for a lower price and face stockout. Here suppliers wait for a higher payment and face assignment competition. In two-sided markets, Chen and Hu (2020) study Poisson arrivals with strategic waiting, and Hu et al. (2022) obtain both high–low and low–high surge paths from rider and driver temporal responses. The present model isolates a single request, an ex ante failure-contingent menu, and the posterior created by universal rejection.

Platform bonuses, dispatch, and strategic rejection. Wu et al. (2022) optimize multistage order-specific bonuses in meal delivery using an acceptance model driven by the current bonus. Their fixed-later-bonus ablation finds no detectable anticipation before that bonus, although the future path is not described as disclosed and committed. This evidence is an external-validity challenge for the strategic channel studied here and makes announcement timing an empirical requirement. Sigg et al. (2025) model workers who reject a delivery request so that it is reoffered at a higher payment. Broadcast, supplier-menu, and notification models also study contention and a second attempt after no response, but vary outreach, assignment, or demand batching rather than solve the same-request announced payment game (Sun et al. 2020, Horner et al. 2021, Ausseil et al. 2022, Feng et al. 2023, Qin et al. 2025).

Latent supply, strategic availability, and market thickness. Chen (2012) derives private-cost sellers' willingness to wait for higher buyer offers when the seller count is fixed and public and the buyer does not precommit. Bai et al. (2023) let a platform announce a supply-contingent price and compensation rule when true supply is random and providers can hide availability. Strategic rejection, relocation, and acceptance are likewise established in ride-hailing and service-platform models (Guda and Subramanian 2019, Garg and Nazerzadeh 2022, Afèche et al. 2023, Feng and Wang 2023, Meskar et al. 2023). Poisson population uncertainty determines our assignment and failure posteriors, but Poisson supply and market thickness are tools rather than standalone contributions.

Market thickness itself has direct theoretical and operational precedents. Loertscher et al. (2022) balance gains from accumulating traders against delay in a dynamic bilateral-trade mechanism. Using online food-delivery operations, Zhao et al. (2024) show how processing time changes the thickness available for matching. Together they establish the broader economics of intermediate and optimally chosen market thickness. Our object holds m fixed and compares two payment-policy classes: $V(m)$ is the completion advantage of optimal rescue pricing over optimal flat pricing in that same environment.

Empirical work has also structurally quantified matching and participant responses in ride-hailing. For example, Zhang et al. (2026) estimate driver acceptance and relocation together with rider choice in a centralized matching system. The empirical implementation here consequently requires variation that disciplines announced same-request rescue, screening history, and rider continuation in addition to driver acceptance and market density.

Against these literatures, we characterize the cutoff-WPBE and completion-design problem created jointly by an ex ante same-request posted-payment menu, latent realized competition, a private rider's post-failure continuation, and a continuation pool containing screened incumbents and potentially unscreened entrants. The global design and thickness theorems are proved for the stated uniform no-entry core; the entry extension is local and its global policy comparisons remain numerical.

3. Institutional Setting and Empirical Requirements (Protocol Only)

This section records the empirical contract that must be satisfied before the model can be called calibrated. The current repository contains no request-level data, data dictionary, announcement protocol, or exposure table. Accordingly, every empirical entry below is marked DATA REQUIRED, and no parameter value in the paper is described as an estimate.

3.1. Operational setting and policy timing

An empirical implementation requires a platform on which one request is shown to a well-defined incumbent response set, both p_1 and the failure-contingent p_2 are credibly disclosed before the first response, and a second response window follows universal rejection. The logs must distinguish three events: an incumbent who remains eligible, a newly eligible entrant, and a driver who was never exposed. They must also show whether the rider abandons, repeats p_1 , or activates p_2 . A no-acceptance record without the corresponding exposure set cannot distinguish universal rejection from zero supply.

The theoretical transfer is paid by the rider to the selected driver and the objective is completion. If the empirical payment is a platform-funded bonus, a fare component, or a bundled transfer, the incidence must be documented and the mapping treated as a reduced-form calibration rather than an exact institutional identity. Profit, subsidy expenditure, and welfare require additional primitives and are not implied by a completion-maximizing model.

3.2. Data construction and measurement map

The empirical unit is a request-response-window record linked to driver-level exposures and rider continuation events. Table 1 is the minimum measurement map. Raw or restricted observations must remain outside Git; only schemas, aggregate outputs, and reproducible code belong in the repository.

The sample-construction table will report the data source, markets, dates, randomization or policy arms, original request count, exclusions, missingness, censoring, and final linked sample. Descriptive evidence will separately show first-window acceptance, universal rejection, rider continuation, rescue activation, terminal acceptance, and completion across thickness cells.

3.3. Identification gates

The announced-rescue channel is empirically identified only if p_2 was known and credible before incumbents chose whether to accept p_1 . Static price variation can discipline current-payment acceptance and the flat benchmark, but cannot by itself identify anticipatory waiting. Endogenous menu assignment requires an instrument or a transparent selection model. Likewise, α and γ cannot be separated without persistent driver identifiers and response-window eligibility records.

Table 1 Model-to-data measurement map (design only; no estimates).

Object	Economic meaning	Required empirical counterpart	Status
p_1	Initial payment	Amount displayed to each side before the first response	DATA REQUIRED
p_2	Committed rescue payment	Amount and disclosure timestamp preceding first-round decisions	DATA REQUIRED
m	Public expected incumbent thickness	Logged supply forecast or pre-decision market-state proxy	DATA REQUIRED
N^I	Latent realized incumbent count	Initial eligible/exposed roster, used through a measurement model	DATA REQUIRED
α	Incumbent survival	Same driver remains eligible after universal rejection	DATA REQUIRED
γm	Fresh supply intensity	Newly eligible second-window drivers not in the incumbent roster	DATA REQUIRED
β	Rider continuation factor	Posting and abandon/repeat/rescue choices across delay and payment variation	DATA REQUIRED
δ	Incumbent waiting factor	Same-driver decisions across windows and waiting-time variation	DATA REQUIRED
F, G	Cost and value distributions	Choice variation with exposure denominators and exogenous payments	DATA REQUIRED

Expected thickness m is not an observed realization of N^I . Each parameter will be classified as measured, estimated, calibrated, or fixed for sensitivity before empirical results are reported.

Until these gates are met, the empirical contribution is a pre-specified design rather than a result. Section 11 states the calibration criterion, validation moments, uncertainty procedure, and counterfactuals that will be populated when the required inputs become available.

4. Model

4.1. Primitives and policy

Table 2 defines the primitives once. The analytical results below impose $F(c) = c$ and $G(v) = v$ on $[0, 1]$; the empirical design allows parametric or semiparametric alternatives and must label that change explicitly.

There is one potential request and two periods. Market thickness $m > 0$ is public. Before rider or driver private information is used, the platform commits to

$$\mathcal{P}^D = \{(p_1, p_2) : 0 \leq p_1 \leq p_2 \leq 1\}.$$

The second payment is available only following first-period failure and cannot be revised ex post. A flat policy is (p, p) . Payment is a transfer from the rider to the selected driver. The platform maximizes the unconditional probability that the potential request is completed; it does not maximize profit or welfare.

The rider has $v \sim U[0, 1]$. A period-1 completion yields $v - p_1$ and a period-2 completion at payment p_j yields $\beta v - p_j$, where $\beta \in (0, 1)$. Incumbents have a separate discount factor $\delta \in (0, 1]$: if an incumbent waits and is selected in period 2, her period-1-equivalent net payoff is $\delta(p_j - c)$. A fresh

Table 2 Model primitives and empirical counterparts.

Symbol	Domain	Economic meaning	Empirical counterpart
p_1, p_2	$0 \leq p_1 \leq p_2 \leq 1$	Initial and committed rescue payments	Displayed menu and timestamps
m	$(0, \infty)$	Public expected incumbent thickness	Pre-decision supply state
N^I	$\text{Pois}(m)$	Latent realized incumbent count	Initial eligible exposure roster
α	$[0, 1]$	Incumbent survival after failure	Linked incumbent eligibility
γ	$[0, \infty)$	Fresh-entry intensity multiplier	New second-window exposure rate
β	$(0, 1)$	Rider continuation factor	Posting and continuation choices
δ	$(0, 1]$	Incumbent waiting factor	Cross-window incumbent decisions
F	cdf on $[0, 1]$	Driver cost distribution	Acceptance by payment and exposure
G	cdf on $[0, 1]$	Rider value distribution	Posting and continuation by payment

The main global analysis uses uniform F and G . Expected thickness m is distinct from the latent realization N^I .

entrant arrives only in period 2 and evaluates the contemporaneous payoff $p_j - c$. Thus β measures rider patience and δ measures incumbent-driver patience; the common-discount specification is the restriction $\delta = \beta$. From the request perspective the incumbent count is

$$N^I \sim \text{Pois}(m).$$

The realization is latent to every player. Incumbent costs are i.i.d. $U[0, 1]$ and remain fixed across periods. An accepting driver incurs her cost only if selected; an unselected acceptor earns and incurs zero.

Conditional on the rider continuing after universal rejection, each incumbent survives independently with probability $\alpha \in [0, 1]$ and a fresh pool

$$N^E \sim \text{Pois}(\gamma m), \quad \gamma \geq 0,$$

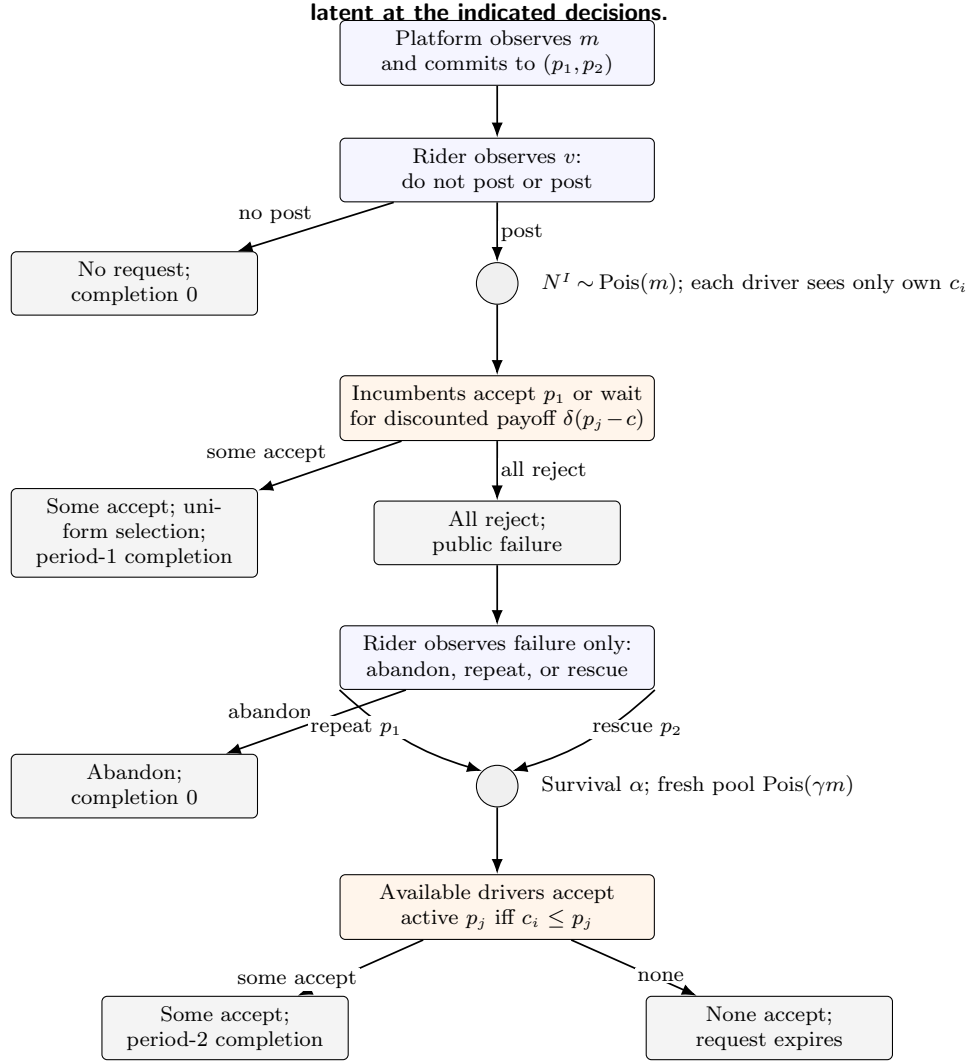
arrives independently with i.i.d. uniform costs. These realizations occur after the rider's abandon-repeat-rescue decision, and the rider observes neither count nor survival. Moving the independent latent draws immediately before that decision would be outcome-equivalent. At the terminal node, an available driver accepts the active payment p_j exactly when $c \leq p_j$.

4.2. Timing and information

Figure 1 gives the reduced extensive form. The driver nodes represent simultaneous Bayesian moves, not observation of a realized pool.

The timing is as follows.

Figure 1 Reduced extensive form. The realized incumbent count, rival costs, survival, and fresh entry remain



Stage 1. The platform observes the public state m and commits to the entire menu (p_1, p_2) . The rescue payment cannot be revised after observing failure.

Stage 2. The rider privately observes v and chooses whether to post the request. A nonposted request yields zero completion.

Stage 3. Conditional on posting, the latent incumbent pool is realized. Each incumbent observes the menu and own cost but not the number or costs of rivals.

Stage 4. Incumbents simultaneously accept p_1 or wait. If at least one accepts, one acceptor is selected uniformly and the request completes in period 1.

Stage 5. If every incumbent rejects, universal rejection becomes public. It reveals the first-period screening event but not the realized pool size.

Stage 6. After observing failure, the rider chooses abandon, repeat p_1 , or activate p_2 .

Stage 7. Each rejected incumbent independently survives with probability α . Independently, a fresh unscreened pool with mean γm enters. Counts, survival realizations, and entrant costs remain latent.

Stage 8. Available drivers accept the active terminal payment when it covers their cost. If at least one accepts, uniform selection produces period-2 completion; otherwise the request expires.

4.3. Equilibrium class, beliefs, and tie-breaking

We use weak perfect Bayesian equilibrium (WPBE), restricted to anonymous, symmetric, pure first-period driver cutoffs. Under cutoff a , incumbents below a accept and those above a wait. The cutoff type accepts at an interior or upper cutoff and rejects at the lower cutoff; this affects only a null cost type.

The rider posts at a zero payoff when $v = p_1$, and does not post below it. After failure, the following priority rule makes continuation single-valued. If the maximal continuation payoff is zero, the rider repeats iff $\beta v - p_1 \geq 0$ and otherwise abandons. If repeat and rescue tie at a strictly positive maximum, the rider chooses rescue. Terminal drivers accept at a zero surplus. These rules matter when repeat has zero coverage: a positive interval of rider types can then be indifferent between repeat and abandon.

For $p_1 < 1$, posting implies the belief $v \mid \{\text{post}\} \sim U[p_1, 1]$. Since supply is independent of value, failure leaves this value posterior unchanged. Poisson splitting determines the rider's supply posterior after failure. For a focal incumbent, Palm conditioning leaves the number of rivals distributed as $\text{Pois}(m)$. At $(p_1, p_2) = (1, 1)$, posting is a null event; we complete the off-path driver subgame with cutoff one and define completion as zero.

5. Equilibrium under an arbitrary announced menu

Fix a candidate cutoff $a \in [0, p_1]$ and define

$$\phi(x) = \begin{cases} (1 - e^{-x})/x, & x > 0, \\ 1, & x = 0, \end{cases} \quad \lambda_j(a) = m\{\alpha(p_j - a) + \gamma p_j\}, \quad C_j(a) = 1 - e^{-\lambda_j(a)}. \quad (1)$$

LEMMA 1 (Poisson competition and failure posterior). *Under cutoff a , first-period completion conditional on posting is $1 - e^{-ma}$. A focal acceptor's expected assignment share is $\phi(ma)$, while a focal waiting driver reaches failure against all rivals with probability e^{-ma} . Conditional on universal rejection, incumbents above a remain a Poisson marked population. After survival and fresh entry, terminal coverage at p_j is $C_j(a)$, and an eligible focal driver's terminal assignment share is $\phi(\lambda_j(a))$.*

The assignment identity follows from

$$\mathbb{E} \left[\frac{1}{1+Z} \right] = \int_0^1 \mathbb{E}[t^Z] dt = \frac{1 - e^{-x}}{x}, \quad Z \sim \text{Pois}(x).$$

Poisson marking gives independent counts below and above a . Conditional on no point below a , the above- a process is unchanged. This is not the law of $N^I - 1$ conditional on $N^I \geq 1$.

After failure, rider payoffs from abandon, repeat, and rescue are

$$0, \quad C_1(a)(\beta v - p_1), \quad C_2(a)(\beta v - p_2).$$

If $C_2 > C_1$, define

$$v^M(a) = \frac{C_2(a)p_2 - C_1(a)p_1}{\beta[C_2(a) - C_1(a)]};$$

if $C_2 = C_1$, put $v^M = +\infty$. Conditional on posting, the repeat and rescue masses are

$$\eta^R(a) = \frac{[\min\{v^M(a), 1\} - p_1/\beta]^+}{1 - p_1}, \quad \eta(a) = \frac{[1 - v^M(a)]^+}{1 - p_1}, \quad (2)$$

and

$$\eta^R(a) + \eta(a) = \rho(p_1) := \frac{[1 - p_1/\beta]^+}{1 - p_1}. \quad (3)$$

The initial posting payoff is

$$(1 - e^{-ma})(v - p_1) + e^{-ma} \max\{0, C_1(\beta v - p_1), C_2(\beta v - p_2)\},$$

so the rider posts exactly when $v \geq p_1$.

A focal incumbent of cost c obtains

$$A(c; a) = \phi(ma)(p_1 - c) \quad (4)$$

from accepting immediately and

$$W(c; a) = \delta \alpha e^{-ma} \{ \eta^R(a) \phi(\lambda_1(a)) [p_1 - c]^+ + \eta(a) \phi(\lambda_2(a)) [p_2 - c]^+ \} \quad (5)$$

from waiting. These are conditional on observing a posted request; the posting mass must not multiply a driver's sequential payoff.

For $c \leq p_1$, the margin $A - W$ is affine and strictly decreasing in c :

$$A(c; a) - W(c; a) = B(a)(p_1 - c) - D(a), \quad (6)$$

where

$$B(a) = \phi(ma) - \delta \alpha e^{-ma} \{ \eta^R(a) \phi(\lambda_1) + \eta(a) \phi(\lambda_2) \},$$

$$D(a) = \delta \alpha e^{-ma} \eta(a) \phi(\lambda_2) (p_2 - p_1).$$

Indeed, $B(a) > 0$ for $p_1 > 0$ because

$$B(a) \geq \phi(ma) - \delta e^{-ma} \geq 0,$$

with strictness from $a > 0$ or, at $a = 0$, from $\rho(p_1) < 1$. Hence a best response is genuinely a cutoff.

Define the cutoff-type margin

$$f_m(a; p_1, p_2) = \phi(ma)(p_1 - a) - \delta \alpha e^{-ma} \{ \eta^R(a) \phi(\lambda_1)(p_1 - a) + \eta(a) \phi(\lambda_2)(p_2 - a) \}. \quad (7)$$

PROPOSITION 1 (Symmetric cutoff WPBE). *For $0 < p_1 < 1$, the equilibrium cutoff set is*

$$\begin{aligned} \mathcal{E}_m^{\alpha, \delta, \gamma}(p_1, p_2) = & \{0 : f_m(0; p_1, p_2) \leq 0\} \\ & \cup \{a \in (0, p_1) : f_m(a; p_1, p_2) = 0\} \\ & \cup \{p_1 : f_m(p_1; p_1, p_2) \geq 0\}. \end{aligned} \quad (8)$$

It is nonempty and compact. At $p_1 = 0$, the only aggregate cutoff is zero with the lower-boundary reject action; at $(1, 1)$, set $\mathcal{E}_m^{\alpha, \delta, \gamma}(1, 1) = \{1\}$. Thus every feasible menu admits an anonymous symmetric pure-cutoff WPBE.

At the upper boundary,

$$f_m(p_1; p_1, p_2) = -\delta \alpha e^{-mp_1} \eta(p_1) \phi(\lambda_2(p_1))(p_2 - p_1) \leq 0,$$

so $a = p_1$ is eligible precisely when the right side is zero. Continuity of f_m , the lower-boundary test, and the intermediate value theorem give existence; endpoint limits remain eligible, proving compactness.

For $p_1 < 1$, completion on cutoff branch a is

$$M_m^{\alpha, \gamma}(p_1, p_2; a) = (1 - p_1) [1 - e^{-ma} + e^{-ma} \{ \eta^R(a) C_1(a) + \eta(a) C_2(a) \}]. \quad (9)$$

We define the conservative value within the stated class by

$$\underline{M}_m^{\alpha, \delta, \gamma}(p_1, p_2) = \min_{a \in \mathcal{E}_m^{\alpha, \delta, \gamma}(p_1, p_2)} M_m^{\alpha, \gamma}(p_1, p_2; a), \quad (10)$$

which is well-defined by Proposition 1. The announced value is

$$D_{\alpha, \delta, \gamma}^*(m) = \sup_{(p_1, p_2) \in \mathcal{P}^D} \underline{M}_m^{\alpha, \delta, \gamma}(p_1, p_2). \quad (11)$$

We retain a supremum here; attainment will be proved for the main no-entry baseline rather than assumed in the general model.

6. Flat-payment benchmark

PROPOSITION 2 (Flat policy). *For every $m > 0$ and $p > 0$, (p, p) has the unique numeric cutoff $a = p$. Its completion is*

$$Q_\gamma^F(m, p) = (1 - p)(1 - e^{-mp}) + e^{-mp}[1 - p/\beta]^+(1 - e^{-\gamma mp}). \quad (12)$$

Consequently

$$F_\gamma^*(m) = \max_{p \in [0, 1]} Q_\gamma^F(m, p) \quad (13)$$

is attained.

For $a < p$, the flat cutoff margin factors as

$$(p - a) [\phi(ma) - \delta \alpha e^{-ma} \rho(p) \phi(m\{\alpha(p - a) + \gamma p\})] > 0,$$

so no lower cutoff is sustainable. At $a = p$, surviving rejectors have cost above p and do not accept the same payment later; only fresh drivers can complete after failure, which gives (12). Continuity on $[0, 1]$ gives the maximum in (13).

Let

$$\mathcal{P}_\gamma^F(m) = \arg \max_{p \in [0, 1]} Q_\gamma^F(m, p) \quad \text{and} \quad p^0(m) = \frac{1 + m - W_0(e^{1+m})}{m}. \quad (\text{F1})$$

Here W_0 is the principal real branch of the Lambert W function, and $p^0(m)$ is the unique maximizer of the first-window objective $q_0(m, p) = (1 - p)(1 - e^{-mp})$.

LEMMA 2 (Geometry of optimized flat pricing). *For $m > 0$, $\beta \in (0, 1)$, and $\gamma > 0$,*

$$\mathcal{P}_\gamma^F(m) \subset (0, \beta) \cup \{p^0(m)\}, \quad 0 < p^0(m) < \frac{1}{2}. \quad (\text{F2})$$

The point $p^0(m)$ can be optimal only when $\beta < p^0(m)$. Hence, if $\beta \geq p^0(m)$ —in particular, if $\beta \geq 1/2$ —every optimized flat price is strictly below β . The correspondence $\mathcal{P}_\gamma^F(m)$ need not be single valued.

The lemma prevents a hidden uniqueness assumption in the mixed-entry result below. When $\beta < p^0(m)$, a low-price maximizer and $p^0(m)$ can tie for some $\gamma > 0$; the theorem therefore requires one active optimizer, not a unique optimizer.

7. Extension: Surviving Incumbents and Fresh Driver Entry

After universal rejection, two economically distinct sources of terminal supply coexist. A surviving incumbent appears with probability α , has already revealed a cost above the first-period cutoff, and discounts the payoff from waiting by δ . A fresh entrant arrives at intensity γm , has not

been screened by the first response, and evaluates the terminal payment contemporaneously. The resulting eligible intensity at p_j is

$$m\{\alpha(p_j - a) + \gamma p_j\},$$

which records screening history rather than collapsing continuation supply to the single total $(\alpha + \gamma)m$. This composition effect is the focus of the extension. The central mixed-continuation case has $0 < \alpha < 1$ and $\gamma > 0$: some rejected incumbents disappear, some remain, and new drivers enter before the terminal response window. The cases $\gamma = 0$ or $\alpha \in \{0, 1\}$ are nested boundaries, not substitutes for the mixed case.

The general equilibrium characterization permits a sharp local comparison with both sources present. Fix $0 < p < \beta$ and abbreviate

$$x = mp, \quad R = e^{-x}, \quad E = e^{-\gamma x}, \quad \sigma = \phi(x), \quad \ell = \phi(\gamma x), \quad \rho = \frac{1 - p/\beta}{1 - p}. \quad (\text{L1})$$

For $\alpha + \gamma > 0$, define the limiting repeat-rescue switch and rescue mass by

$$\bar{v} = \frac{1}{\beta} \left[p + \frac{1 - E}{mE(\alpha + \gamma)} \right], \quad \eta^0 = \frac{1 - \bar{v}}{1 - p}. \quad (\text{L2})$$

At $\gamma = 0$, the fraction in (L2) is zero.

THEOREM 1 (Fresh-entry local rescue). *Suppose $m > 0$, $0 < p < \beta < 1$, $0 < \delta \leq 1$, $0 \leq \alpha \leq 1$, $\gamma \geq 0$, $\alpha + \gamma > 0$, and $\bar{v} < 1$. For all sufficiently small $\varepsilon > 0$, the menu $(p, p + \varepsilon)$ has a unique anonymous symmetric cutoff WPBE, with*

$$a_\varepsilon = p - \kappa \varepsilon + o(\varepsilon), \quad \kappa = \frac{\delta R \alpha \ell \eta^0}{\sigma - \delta R \alpha \rho \ell}. \quad (\text{L3})$$

Every candidate cutoff is uniformly localized by

$$0 \leq p - a_\varepsilon \leq \frac{\delta \alpha \rho}{1 - \delta \alpha \rho} \varepsilon. \quad (\text{L4})$$

The right derivative of completion is

$$\left. \frac{dM_m^{\alpha, \gamma}(p, p + \varepsilon; a_\varepsilon)}{d\varepsilon} \right|_{0+} = m(1 - p)R\eta^0 \frac{T_\delta}{\sigma - \delta R \alpha \rho \ell} > 0, \quad (\text{L5})$$

where

$$T_\delta = E(\alpha + \gamma)\sigma - \delta R \alpha \ell \{1 - \rho + \rho E(1 + \gamma)\} > 0. \quad (\text{L6})$$

The activity condition $\bar{v} < 1$ says that a positive mass of riders uses marginal rescue after failure. It is automatic at $\gamma = 0$ but substantive with entry. The knife edge $\bar{v} = 1$ needs a second-order rider-mass expansion and is not included. For $\gamma > 1$, activity cannot simply be deleted from the sign claim: for example

$$x = 1, \quad p = \frac{1}{2}, \quad m = 2, \quad \beta = \frac{51}{100}, \quad \alpha = 1, \quad \gamma = 10, \quad \delta = \frac{1}{2}$$

has $\bar{v} > 1$ and makes T_δ in (L6) negative. Thus the theorem's restriction is economically and mathematically real.

For $p < \beta$, define the activity index

$$\mathcal{A}_{\alpha,\gamma}(p) = me^{-\gamma mp}(\alpha + \gamma)(\beta - p) - (1 - e^{-\gamma mp}). \quad (\text{L7})$$

Then $\mathcal{A}_{\alpha,\gamma}(p) > 0$ is equivalent to $\bar{v} < 1$, or equivalently

$$\alpha > \frac{e^{\gamma mp} - 1}{m(\beta - p)} - \gamma. \quad (\text{L8})$$

THEOREM 2 (Optimized-flat dominance with mixed continuation supply). *Fix $m > 0$, $\beta \in (0, 1)$, $\delta \in (0, 1]$, $0 < \alpha < 1$, and $\gamma > 0$. Suppose that some $p^* \in \mathcal{P}_\gamma^F(m)$ satisfies*

$$p^* < \beta, \quad \mathcal{A}_{\alpha,\gamma}(p^*) > 0. \quad (\text{L9})$$

For every sufficiently small $\varepsilon > 0$, the menu $(p^, p^* + \varepsilon)$ has a unique anonymous symmetric pure-cutoff WPBE and*

$$\underline{M}_m^{\alpha,\delta,\gamma}(p^*, p^* + \varepsilon) = F_\gamma^*(m) + \Lambda(p^*)\varepsilon + o(\varepsilon), \quad \Lambda(p^*) > 0. \quad (\text{L10})$$

Consequently,

$$D_{\alpha,\delta,\gamma}^*(m) > F_\gamma^*(m). \quad (\text{L11})$$

Neither uniqueness of the flat optimizer nor attainment of the general dynamic-policy supremum is required.

If instead $p < \beta$ and $\mathcal{A}_{\alpha,\gamma}(p) < 0$, then for every sufficiently small $\varepsilon > 0$ the same-price perturbation $(p, p + \varepsilon)$ has the unique cutoff $a = p$, no rider activates rescue, and its completion equals $Q_\gamma^F(m, p)$ exactly. No first-order claim is made at the knife edge $\mathcal{A}_{\alpha,\gamma}(p) = 0$.

The coefficient $\Lambda(p^*)$ is the expression in (L5), evaluated at $p = p^*$. Condition (L9) has a direct behavioral interpretation: after flat failure, even the highest rider type must strictly prefer access to the marginally broader rescue pool over repeating p^* . More fresh entrants do not make this condition automatic, because entry also raises repeat coverage.

Two different quantities can reasonably be called “the amount” of continuation supply. Conditional on universal rejection at the flat cutoff $a = p$, the expected number of second-window potential drivers is

$$m\tau_p, \quad \tau_p = \alpha(1 - p) + \gamma. \quad (\text{L12})$$

By contrast, holding the cutoff fixed, the number made newly eligible by a small payment increase has slope

$$\left. \frac{1}{m} \frac{\partial \lambda_2}{\partial p_2} \right|_{a=p} = \alpha + \gamma. \quad (\text{L13})$$

The next result fixes each measure in turn at the flat benchmark. These are local comparisons, not equal-baseline optimized-policy rankings. Along the equilibrium response $a_\varepsilon = p - \kappa\varepsilon + o(\varepsilon)$, raw headcount changes at rate $m\alpha\kappa$, and terminal eligible intensity changes at rate $m\{\alpha(1 + \kappa) + \gamma\}$. The composition response is precisely why neither baseline normalization removes strategic waiting.

THEOREM 3 (Supply composition and strategic waiting). *Fix $m > 0$, $0 < p < \beta < 1$, and $\delta \in (0, 1]$, and consider $(p, p + \varepsilon)$.*

(i) *Fixed continuation headcount. Fix $h > 0$ and set*

$$\gamma_h(\alpha) = h - (1 - p)\alpha, \quad A_h(\alpha) = \alpha + \gamma_h(\alpha) = h + p\alpha \quad (\text{L14})$$

for $0 \leq \alpha \leq \min\{1, h/(1 - p)\}$. This holds the conditional headcount in (L12) fixed at mh . Define

$$\ell_h(\alpha) = \phi(\gamma_h(\alpha)mp), \quad \eta_h(\alpha) = \frac{1}{1 - p} \left[1 - \frac{1}{\beta} \left\{ p + \frac{e^{\gamma_h(\alpha)mp} - 1}{mA_h(\alpha)} \right\} \right]. \quad (\text{L15})$$

On every interval on which $\eta_h(\alpha) > 0$, the unique local cutoff is

$$a_\varepsilon(\alpha) = p - \kappa_h(\alpha)\varepsilon + o(\varepsilon), \quad \kappa_h(\alpha) = \frac{\delta e^{-mp}\alpha\ell_h(\alpha)\eta_h(\alpha)}{\phi(mp) - \delta e^{-mp}\alpha\rho(p)\ell_h(\alpha)}, \quad (\text{L16})$$

and $\kappa_h(\alpha)$ is strictly increasing in α .

(ii) *Fixed price-marginal capacity. Fix $s > 0$ and set*

$$\gamma_s(\alpha) = s - \alpha, \quad 0 \leq \alpha \leq \min\{1, s\}. \quad (\text{L17})$$

This holds (L13) fixed at s . With

$$\ell_s(\alpha) = \phi((s - \alpha)mp), \quad \eta_s(\alpha) = \frac{1}{1 - p} \left[1 - \frac{1}{\beta} \left\{ p + \frac{e^{(s - \alpha)mp} - 1}{ms} \right\} \right], \quad (\text{L18})$$

the analogous coefficient

$$\kappa_s(\alpha) = \frac{\delta e^{-mp}\alpha\ell_s(\alpha)\eta_s(\alpha)}{\phi(mp) - \delta e^{-mp}\alpha\rho(p)\ell_s(\alpha)} \quad (\text{L19})$$

is strictly increasing on the active set $\eta_s(\alpha) > 0$.

Thus, within the active local-rescue region, replacing unscreened entrants with screened surviving incumbents strictly increases the first-order acceptance-erosion coefficient under either fixed-supply comparison.

The composition effect on completion is subtler than the cutoff effect.

THEOREM 4 (Closed-form endpoint boundary for local rescue value). *Fix $0 < s \leq 1$, impose $\alpha + \gamma = s$, and suppose the all-entrant composition is active:*

$$p + \frac{e^{smp} - 1}{ms} < \beta. \quad (\text{L20})$$

Let $L_s(\alpha)$ denote the right derivative in (L5) at composition $(\alpha, s - \alpha)$. Write

$$x = mp, \quad R = e^{-x}, \quad E_s = e^{-sx}, \quad \sigma = \phi(x), \quad \rho = \frac{1 - p/\beta}{1 - p},$$

$$\eta_E = \frac{1}{1 - p} \left[1 - \frac{1}{\beta} \left\{ p + \frac{e^{sx} - 1}{ms} \right\} \right], \quad H = \frac{E_s \eta_E}{\rho}. \quad (\text{L21})$$

Then $0 < H < 1$ and the all-entrant and all-incumbent values are

$$L_s(0) = m(1 - p)RsE_s\eta_E, \quad L_s(s) = m(1 - p)Rs\rho \frac{\sigma - \delta R}{\sigma - \delta Rs\rho}. \quad (\text{L22})$$

Define

$$\delta_c(m, p, \beta, s) = \frac{\sigma(1 - H)}{R(1 - Hs\rho)}. \quad (\text{L23})$$

For $\delta \in (0, 1]$,

$$\begin{cases} L_s(s) > L_s(0), & \delta < \delta_c(m, p, \beta, s), \\ L_s(s) = L_s(0), & \delta = \delta_c(m, p, \beta, s), \\ L_s(s) < L_s(0), & \delta > \delta_c(m, p, \beta, s). \end{cases} \quad (\text{L24})$$

Hence the two pure compositions have generically different marginal completion values, but their ranking depends on incumbent patience; $L_s(\alpha)$ need not be monotone between the endpoints.

The nonmonotonicity is not a logical possibility only. At $(m, p, \beta, \delta, s) = (1, 0.1, 0.5, 1, 1)$, direct evaluation of the exact local formula at three strictly mixed compositions gives approximately

$$L_s(0.75) = 0.24267, \quad L_s(0.92) = 0.20589, \quad L_s(0.99) = 0.22333. \quad (\text{L25})$$

Thus local completion value first falls and then rises with the incumbent share even though cut-off erosion rises throughout. This counterexample and the activity boundary are included in the numerical falsification tests.

8. Global menu design without fresh entry

We now solve the primary baseline

$$\gamma = 0, \quad m > 0, \quad \alpha \in (0, 1], \quad \beta \in (0, 1), \quad \delta \in (0, 1].$$

Let $k = \alpha m$. Menus with $p_1 \geq \beta$ or $p_2 \geq \beta$ cannot induce positive-mass rescue and are outcome-equivalent to the flat first payment. The nontrivial region is therefore $0 \leq p_1 < p_2 < \beta$.

For $a \in [0, p_1]$, put

$$C(x) = 1 - e^{-kx}, \quad h(a) = \begin{cases} (e^{ma} - 1)/a, & a > 0, \\ m, & a = 0, \end{cases} \quad (\text{14})$$

and

$$A_z(a) = (\beta - z)C(z - a), \quad z \in \{p_1, p_2\}, \quad B_1 = \beta(1 - p_1). \quad (\text{15})$$

The repeat-rescue switch lies below one exactly when $A_{p_2}(a) > A_{p_1}(a)$. Substitution of the switch type into continuation coverage gives the exact cancellation

$$K_{12}(a) = \frac{\max\{A_{p_1}(a), A_{p_2}(a)\}}{B_1}. \quad (16)$$

Multiplying the cutoff driver's payoff margin by the positive factor me^{ma} gives the sign-equivalent scalar

$$\mathcal{D}_{12}^\delta(a) = (p_1 - a)h(a) - \delta K_{12}(a). \quad (17)$$

The repeat branch cannot contain a cutoff below p_1 :

$$(p_1 - a)h(a) > \frac{A_{p_1}(a)}{B_1} \geq \delta \frac{A_{p_1}(a)}{B_1}, \quad 0 \leq a < p_1. \quad (18)$$

For $a > 0$, this is $\phi(ma) > e^{-ma}$ combined with $\alpha\rho(p_1)\phi(k(p_1 - a)) \leq 1$; at $a = 0$, strictness follows from $\rho(p_1) < 1$. Hence every zero of (17) under a strict active menu lies on the rescue branch, not at the rider kink.

THEOREM 5 (Global cutoff uniqueness). *Fix $0 \leq p_1 < p_2 < \beta$. If $\mathcal{D}_{12}^\delta(0) \leq 0$, the unique numeric cutoff is the lower-boundary cutoff $a = 0$. If $\mathcal{D}_{12}^\delta(0) > 0$, there is a unique cutoff $a \in (0, p_1)$. Equality at zero never coexists with a positive root. Together with the flat, repeat-only, $p_1 = 0$, and $\alpha = 0$ boundary cases, every no-entry menu has a unique numeric symmetric cutoff under the stated cutoff-type convention.*

The one-crossing argument is global. At an active-rescue zero write $u = p_1 - a$, $z = p_2 - a$, $b = \beta - p_2$, and $B = \beta(1 - p_1)$. The root equation is

$$uh(a) = \delta \frac{b}{B} C(z).$$

Since $B - b = p_2 - \beta p_1 > 0$ and $h(a) \geq m$, $u < C(z)/m$. For $R(a) = uh(a)/C(z)$,

$$\frac{R'(a)}{R(a)} = -\frac{1}{u} + \frac{h'(a)}{h(a)} + \frac{k}{e^{kz} - 1} < 0, \quad (19)$$

because

$$\frac{1}{u} - \frac{k}{e^{kz} - 1} > m \frac{1 - \alpha e^{-kz}}{1 - e^{-kz}} \geq m > \frac{h'(a)}{h(a)}.$$

Thus every zero crosses downward. Since $\mathcal{D}_{12}^\delta(p_1) < 0$, two zeros would require an intervening upward crossing. At $p_1 = 0$, the numeric cutoff is zero and the cost-zero type uses the lower-boundary reject action.

8.1. Completion identity and comparison with the same flat payment

At any positive cutoff, indifference gives

$$e^{-ma}K_{12}(a) = \frac{m\phi(ma)(p_1 - a)}{\delta}.$$

Using $1 - e^{-ma} = ma\phi(ma)$ in (9) yields

$$M_m^{\alpha, \delta, 0}(p_1, p_2; a) = (1 - p_1)m\phi(ma) \left[a + \frac{p_1 - a}{\delta} \right], \quad a > 0. \quad (20)$$

At a reject-all equilibrium,

$$M_m^{\alpha, \delta, 0}(p_1, p_2; 0) = \left(1 - \frac{p_2}{\beta} \right) (1 - e^{-kp_2}) \geq \frac{(1 - p_1)mp_1}{\delta}, \quad (21)$$

with equality exactly at the lower-boundary indifference condition. Repeat may be used by positive mass in (21); its contribution cancels algebraically.

Both $\phi(ma)$ and $a + (p_1 - a)/\delta$ weakly decrease in $a > 0$, and the first decreases strictly, so (20) decreases with the cutoff. Theorem 5 then gives a particularly strong comparison. For every $0 < p_1 < p_2 < \beta$, the strict-menu cutoff is below p_1 , and therefore

$$\underline{M}_m^{\alpha, \delta, 0}(p_1, p_2) > F_m(p_1) = (1 - p_1)(1 - e^{-mp_1}). \quad (22)$$

This is a global same- p_1 result, not merely a derivative along one selected branch.

For completeness, a one-sided local expansion at (p_1, p_1) is

$$a_\varepsilon = p_1 - \kappa\varepsilon + o(\varepsilon), \quad \kappa = \frac{\delta e^{-mp_1}\alpha\rho(p_1)}{\phi(mp_1) - \delta e^{-mp_1}\alpha\rho(p_1)}, \quad (23)$$

and

$$\left. \frac{dM(p_1, p_1 + \varepsilon)}{d\varepsilon} \right|_{0+} = \frac{m(1 - p_1)e^{-mp_1}\alpha\rho(p_1)[\phi(mp_1) - \delta e^{-mp_1}]}{\phi(mp_1) - \delta e^{-mp_1}\alpha\rho(p_1)} > 0. \quad (24)$$

The wedge decomposes as $\phi(mp_1) - \delta e^{-mp_1} = [\phi(mp_1) - e^{-mp_1}] + (1 - \delta)e^{-mp_1}$: assignment competition is the first term and incumbent impatience the second.

8.2. The fixed-first-payment problem

Following the design sequence, fix p_1 before optimizing p_2 . For a target cutoff $a < \beta$, maximize

$$A(a, p_2) = (\beta - p_2)[1 - e^{-k(p_2 - a)}], \quad p_2 \in [a, \beta]. \quad (25)$$

It is strictly concave. Its unique maximizer $Q(a)$ satisfies

$$e^{k(Q(a) - a)} - 1 = k(\beta - Q(a)), \quad (26)$$

or

$$Q(a) = \beta - \frac{W_0(e^{1+k(\beta-a)}) - 1}{k}. \quad (27)$$

Moreover

$$a < Q(a) < \frac{a + \beta}{2}.$$

Let

$$S(a) = A(a, Q(a)), \quad S(a) = \frac{k[\beta - Q(a)]^2}{1 + k[\beta - Q(a)]}. \quad (28)$$

Define $Q_0 = Q(0)$, $S_0 = S(0)$, and

$$p_z = \frac{1 - \sqrt{1 - 4\delta S_0/(\beta m)}}{2}. \quad (29)$$

The strict flat inequality at $p_1 = Q_0, a = 0$ gives $0 < p_z < Q_0 < 1/2$.

THEOREM 6 (Optimal rescue conditional on the first payment). *Let*

$$H_m(p_1) = \max_{p_2 \in [p_1, 1]} M_m^{\alpha, \delta, 0}(p_1, p_2).$$

Then:

- (i) *If $0 \leq p_1 \leq p_z$, the unique optimal rescue is $p_2 = Q_0$, the cutoff is the lower-boundary reject-all cutoff $a = 0$, and*

$$H_m(p_1) = S_0/\beta.$$

- (ii) *If $p_z < p_1 < \beta$, there is a unique $a^*(p_1) \in (0, p_1)$ satisfying*

$$(p_1 - a)h(a) = \frac{\delta S(a)}{\beta(1 - p_1)}. \quad (30)$$

The unique optimal rescue is $p_2 = Q(a^(p_1))$, and*

$$H_m(p_1) = (1 - p_1)m\phi(ma^*(p_1)) \left[a^*(p_1) + \frac{p_1 - a^*(p_1)}{\delta} \right].$$

- (iii) *If $\beta \leq p_1 \leq 1$, every $p_2 \in [p_1, 1]$ is outcome-equivalent and*

$$H_m(p_1) = (1 - p_1)(1 - e^{-mp_1}).$$

Thus the conditional supremum is a maximum in every regime.

To see implementability, a positive target cutoff solves

$$A(a, p_2) = \frac{\beta(1 - p_1)(p_1 - a)h(a)}{\delta}. \quad (31)$$

The right side strictly exceeds $A(a, p_1)$ by (18), so (31) cannot generate a repeat-only or kink pseudo-solution. If the envelope is below, equal to, or above the right side, there are respectively zero, one tangent, or two strict implementing prices. Theorem 5 makes each implemented cutoff the unique equilibrium. Completion falls with the cutoff, so the smallest implementable cutoff, attained at tangency, solves the fixed- p_1 problem.

8.3. One-dimensional global design and attainment

Define

$$R_\delta(a) = \frac{\delta S(a)}{\beta h(a)}.$$

The tangent equation is

$$(1 - p_1)(p_1 - a) = R_\delta(a). \quad (32)$$

Because $Q(a) < (1 + a)/2$ and the left side is increasing in p_1 below that vertex, the unique feasible root is the lower quadratic branch

$$P_\delta(a) = \frac{1 + a - \sqrt{(1 - a)^2 - 4R_\delta(a)}}{2}. \quad (33)$$

The upper branch is not a fixed- p_1 rescue optimum. The map P_δ is continuous and strictly increasing from $[0, \beta]$ onto $[p_z, \beta]$. Endpoint injectivity follows from the same envelope one-crossing argument: $P_\delta(a) = P_\delta(0) = p_z$ for $a > 0$ would give the fixed- $p_1 = p_z$ envelope two zeros.

Extend $P_\delta(\beta) = Q(\beta) = \beta$ and set

$$J_\delta(a) = \begin{cases} (1 - P_\delta(a))m\phi(ma) \left[a + \frac{P_\delta(a) - a}{\delta} \right], & a > 0, \\ mp_z(1 - p_z)/\delta = S_0/\beta, & a = 0. \end{cases} \quad (34)$$

Then $J_\delta(\beta) = (1 - \beta)(1 - e^{-m\beta})$, and the complete outer problem is

$$D_{\alpha, \delta, 0}^*(m) = \max \left\{ \max_{a \in [0, \beta]} J_\delta(a), \max_{p \in [\beta, 1]} (1 - p)(1 - e^{-mp}) \right\}. \quad (35)$$

THEOREM 7 (Policy attainment and optimized-flat dominance). *For every no-entry parameter tuple with $m > 0$ and $\alpha > 0$, the announced policy value in (11) is attained. If $\beta \geq 1/2$, every optimal menu is a strict menu*

$$(p_1^*, p_2^*) = (P_\delta(a^*), Q(a^*)), \quad a^* \in \arg \max_{a \in [0, \beta]} J_\delta(a) \subset (0, \beta), \quad (36)$$

and

$$D_{\alpha, \delta, 0}^*(m) > F_0^*(m). \quad (37)$$

No uniqueness of a^* is asserted.

Equation (35) is a maximum of continuous functions on compact intervals, so it proves attainment directly. In the no-entry flat problem,

$$F_m(p) = (1 - p)(1 - e^{-mp})$$

is strictly concave and its unique optimizer is

$$p_F(m) = \frac{1 + m - W_0(e^{1+m})}{m} < \frac{1}{2}. \quad (38)$$

For $\beta \geq 1/2$, $p_F < \beta$, and (22) applied at p_F proves (37), including the equality case $\beta = 1/2$. The boundary portions of (35) do not beat F_0^* , whereas (37) does, so every global optimizer is interior and strict.

The driver discount also yields a clean comparative static. Write $R_1(a) = S(a)/[\beta h(a)]$, so that $(1 - P_\delta)(P_\delta - a) = \delta R_1(a)$. For $a \in (0, \beta)$,

$$\frac{\partial P_\delta(a)}{\partial \delta} = \frac{R_1(a)}{1 + a - 2P_\delta(a)} > 0, \quad \frac{\partial J_\delta(a)}{\partial \delta} = -(1 - e^{-ma}) \frac{\partial P_\delta(a)}{\partial \delta} < 0. \quad (38a)$$

Thus $D_{\alpha, \delta, 0}^*(m)$ is weakly decreasing in δ : more patient incumbents require a higher first payment to implement the same cutoff and continuation coverage. This is a completion comparative static, not a welfare ranking.

8.4. The full patience region

Fix (m, α, δ) and make patience explicit in $D(m; \beta)$. For a fixed strict menu $p_1 < p_2$, its value equals $F_m(p_1)$ for $\beta \leq p_2$ and is strictly increasing for $\beta > p_2$. Indeed, increasing β raises the active-rescue coefficient in (17), shifts the unique cutoff strictly left (or to zero), and then raises completion by (20)–(21).

The scalar representation is jointly continuous in patience. To use a fixed choice set, write $a = \beta x$. The solution of

$$e^{ky} - 1 = k\{\beta(1 - x) - y\}$$

is continuous in (β, x) . At $x = 0$, $h(0) = m$ and $P_{\beta, \delta}(0) = p_z(\beta)$ has a strictly positive discriminant; at $x = 1$, $P_{\beta, \delta}(\beta) = Q_\beta(\beta) = \beta$ and $J_{\beta, \delta}(\beta) = (1 - \beta)(1 - e^{-m\beta})$. These removable extensions and the elementary maximum theorem make $D(m; \beta)$ continuous.

Define

$$\beta_c(m, \alpha, \delta) = \inf\{\beta \in (0, 1) : D_{\alpha, \delta, 0}^*(m; \beta) > F_0^*(m)\}. \quad (39)$$

THEOREM 8 (Patience threshold). *For $m > 0$, $0 < \alpha \leq 1$, and $0 < \delta \leq 1$, the strict-gain region is the single open interval*

$$D_{\alpha, \delta, 0}^*(m; \beta) > F_0^*(m) \iff \beta > \beta_c(m, \alpha, \delta). \quad (40)$$

Moreover

$$0 < -\frac{1}{m} \log(1 - F_0^*(m)) \leq \beta_c(m, \alpha, \delta) \leq p_F(m) < \frac{1}{2}. \quad (41)$$

For fixed (m, α, δ) , $D(m; \beta) = F_0^*(m)$ through the threshold and is strictly increasing above it. The exact scalar test is

$$\beta > \beta_c(m, \alpha, \delta) \iff \max_{a \in [0, \beta]} J_{\beta, \delta}(a) > F_0^*(m). \quad (42)$$

When $\alpha = 0$, the gain region is empty.

Pointwise monotonicity in (38a) also nests the thresholds:

$$0 < \delta_1 < \delta_2 \leq 1 \implies \beta_c(m, \alpha, \delta_1) \leq \beta_c(m, \alpha, \delta_2). \quad (42a)$$

Strict inequality is not asserted globally.

The lower bound follows because an active menu has $p_2 < \beta$ and completion requires an initial incumbent with cost at most p_2 , giving $M < 1 - e^{-m\beta}$. The upper bound follows from a strict menu at p_F when $\beta > p_F$. If gain holds at one patience level, attainment supplies an optimal strict menu; reusing it at any larger β and applying the fixed menu monotonicity gives strict gain and strict value increase. Continuity makes the gain interval open.

9. Market thickness: topology and endpoint rates

Fix $0 < \alpha \leq 1$, $0 < \beta < 1$, $0 < \delta \leq 1$, and $\gamma = 0$. Holding these primitives and the same thickness m fixed, define

$$M_R^*(m) := D_{\alpha, \delta, 0}^*(m), \quad M_F^*(m) := F_0^*(m), \quad V(m) := M_R^*(m) - M_F^*(m). \quad (43)$$

Thus $V(m)$ is the completion-rate gain from allowing an optimally designed rescue menu instead of an optimally chosen flat payment at the same market thickness. It is not the marginal value of adding drivers; $100V(m)$ is the gain in percentage points. The outer representation (35), rather than a local-menu derivative, is the appropriate object because the optimal menu moves with thickness.

For the thin limit, define on $a \in [0, \beta]$

$$P_0^\delta(a) = \frac{1 + a - \sqrt{(1-a)^2 - \delta\alpha(\beta-a)^2/\beta}}{2}, \quad H_0^\delta(a) = a[1 - P_0^\delta(a)] + \frac{\alpha(\beta-a)^2}{4\beta}, \quad (44)$$

and $G_{\alpha, \beta, \delta} = \max_{a \in [0, \beta]} H_0^\delta(a)$.

THEOREM 9 (Thickness topology and exact endpoint regimes). *For fixed $(\alpha, \beta, \delta) \in (0, 1] \times (0, 1) \times (0, 1]$:*

(i) $M_R^*(m)$, $M_F^*(m)$, and $V(m)$ are continuous on $(0, \infty)$. The scalar cutoff and flat-price argmax correspondences are nonempty, compact valued, and upper hemicontinuous.

(ii) If $\beta > 1/2$ and $\delta < 1$, then

$$V(m) \sim \left(G_{\alpha, \beta, \delta} - \frac{1}{4}\right)m, \quad G_{\alpha, \beta, \delta} > \frac{1}{4}. \quad (45)$$

At the no-discount boundary $\delta = 1$, let $a_0 \in (0, 1/2)$ be the unique solution of

$$\alpha(\beta - a_0)^2 = \beta(1 - 2a_0).$$

Then

$$V(m) \sim \frac{1 - 2a_0}{16}m^2. \quad (46)$$

(iii) If $\beta = 1/2$ and $\delta < 1$, then

$$V(m) \sim \frac{\alpha(1-\delta)}{512\{1-\alpha(1-\delta)/2\}} m^3. \quad (47)$$

At $\delta = 1$, the leading cancellation persists one order longer:

$$V(m) \sim \frac{\alpha}{2048} m^4. \quad (48)$$

(iv) If $\beta < 1/2$, there exists $m_0(\alpha, \beta, \delta) > 0$ such that

$$M_R^*(m) = M_F^*(m), \quad V(m) = 0, \quad 0 < m < m_0(\alpha, \beta, \delta). \quad (49)$$

(v) As $m \rightarrow \infty$,

$$1 - M_R^*(m) \sim \frac{\log \log m}{m}, \quad 1 - M_F^*(m) \sim \frac{\log m}{m}, \quad V(m) \sim \frac{\log m}{m}. \quad (50)$$

For each fixed positive (α, β) , the estimates establishing (50) are uniform over the full policy/cutoff domain and uniformly over $\delta \in (0, 1]$. They are not uniform over primitive sequences with $\alpha \downarrow 0$ or $\beta \downarrow 0$.

Equations (45)–(48) show a singular boundary: the limits $m \downarrow 0$ and $\delta \uparrow 1$ do not commute at the first nonzero order. Even slight incumbent impatience creates a first-order screening gain when $\beta > 1/2$ and a cubic gain at $\beta = 1/2$; exact patience removes those terms.

Here is the proof architecture. Let $k = \alpha m$ and let $z_m(a)$ solve

$$z_m(a) + \log[1 + z_m(a)] = k(\beta - a), \quad y_m(a) = z_m(a)/k.$$

Then

$$Q_m(a) = \beta - y_m(a), \quad T_m(a) = \frac{k y_m(a)^2}{\beta[1 + k y_m(a)]}, \quad B_m^\delta(a) = \delta \frac{a}{e^{ma} - 1} T_m(a), \quad (51)$$

with $a/(e^{ma} - 1) = 1/m$ at zero, and

$$P_m^\delta(a) = \frac{1 + a - \sqrt{(1-a)^2 - 4B_m^\delta(a)}}{2}. \quad (52)$$

The scalar objective admits the useful exact form

$$J_m^\delta(a) = [1 - P_m^\delta(a)](1 - e^{-ma}) + T_m(a)e^{-ma}. \quad (53)$$

The strict envelope bound keeps the discriminant positive. Joint continuity and Berge's maximum theorem prove part (i).

Uniformly on $[0, \beta]$, $P_m^\delta \rightarrow P_0^\delta$ and $J_m^\delta/m \rightarrow H_0^\delta$. If $\beta > 1/2$ and $\delta < 1$, the unique $a \in (0, 1/2)$ satisfying $P_0^\delta(a) = 1/2$ gives

$$H_0^\delta(a) = \frac{1}{2} \left[a + \frac{1/2 - a}{\delta} \right] > \frac{1}{4},$$

which proves the strict coefficient in (45); uniform convergence gives its exact value as the maximum in (44). For $\delta = 1$, the first-order term ties the flat benchmark, and the uniform second-order argmax expansion gives (46).

At $\beta = 1/2$, every thin optimizer localizes as $a = 1/2 - cm$. Put $A_\delta = 1 - \alpha(1 - \delta)/2$. Uniformly for bounded c ,

$$J_m^\delta(1/2 - cm) = \frac{m}{4} - \frac{m^2}{16} + m^3 \left\{ \frac{1}{96} + \frac{c}{8} - A_\delta c^2 \right\} + O(m^4). \quad (54)$$

The maximizing rescaled cutoff is $c^* = 1/(16A_\delta)$, while the flat m^3 coefficient is $11/768$. Their difference is exactly the coefficient in (47). When $\delta = 1$ it vanishes; the fourth-order expansion yields (48). In particular, uniformly for bounded c ,

$$\begin{aligned} J_m^1(1/2 - cm) &= \frac{m}{4} - \frac{m^2}{16} + m^3 f_2(c) + m^4 f_3(c) + o(m^4), \\ f_2(c) &= \frac{11}{768} - \left(c - \frac{1}{16} \right)^2, \\ f_3(c) &= -\frac{1}{768} - \frac{c}{24} + \frac{c^2}{4} + 2\alpha c^3, \end{aligned} \quad (54a)$$

whereas

$$F_0^*(m) = \frac{m}{4} - \frac{m^2}{16} + \frac{11}{768}m^3 - \frac{3}{1024}m^4 + o(m^4). \quad (54b)$$

The strict quadratic loss in f_2 and local Lipschitz control of f_3 evaluate the perturbed maximum at $c = 1/16$ to fourth order, and $f_3(1/16) = -3/1024 + \alpha/2048$. Subtracting (54b) proves (48). If $\beta < 1/2$, the bound

$$H_0^\delta(a) \leq a(1 - a) + \frac{(\beta - a)^2}{4\beta} \leq \beta(1 - \beta) < \frac{1}{4}$$

is uniform on $[0, \beta]$. Since the flat optimizer converges to $1/2$, the flat policy wins exactly on a nonempty initial interval, proving (49).

For the thick limit put $x = ma$, $p_1 = P_m^\delta(a)$, and $T = T_m(a)$. Equation (53) gives the same exact nonnegative loss decomposition for every fixed $\delta > 0$:

$$1 - J_m^\delta(a) = p_1(1 - e^{-x}) + (1 - T)e^{-x}. \quad (55)$$

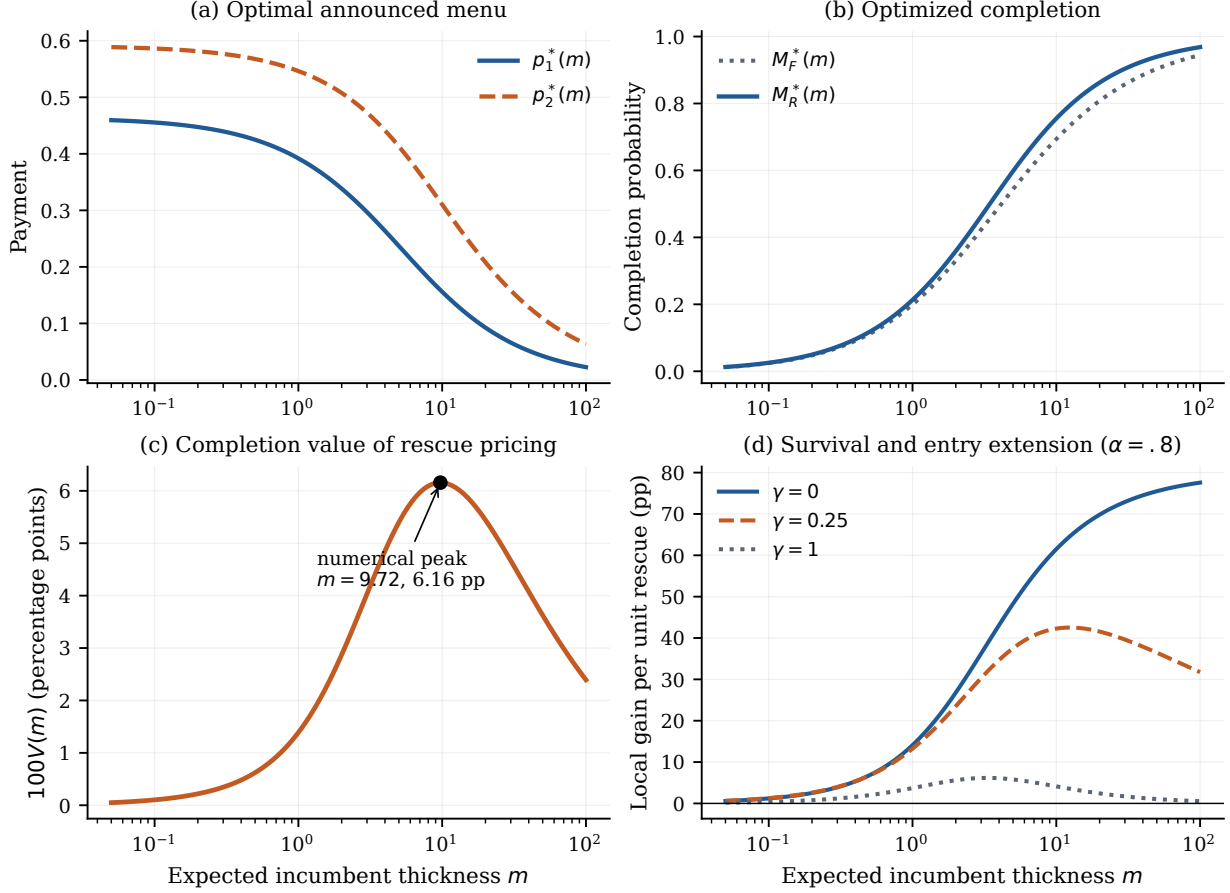
The feasible choice $x = \log \log m$ and the policy-uniform lower bound obtained by splitting at $Q_m(a) = \beta/2$ give the first rate in (50). The exact flat first-order condition gives the second, and subtraction gives the third.

COROLLARY 1 (A best finite thickness, without a shape theorem). *For every fixed positive (α, β, δ) in the maintained ranges, there exists $m^* \in (0, \infty)$ such that*

$$V(m^*) = \max_{m > 0} V(m) > 0. \quad (56)$$

Strict single-peakedness is false when $\beta < 1/2$ because of (49). For $\beta \geq 1/2$, strict single-peakedness remains open for every fixed $\delta \in (0, 1]$.

Figure 2 Theoretical rescue profiles and the survival–entry extension. Panels (a)–(c) use $\alpha = 1$, $\beta = \delta = 0.8$, and $\gamma = 0$. The marked point in panel (c) is a numerical maximizer, not a uniqueness theorem. Panel (d) fixes $\alpha = 0.8$, evaluates a marginal announced rescue at the no-entry flat benchmark, and varies fresh-entry intensity γ . The dashed and dotted profiles have $0 < \alpha < 1$ and $\gamma > 0$, so the terminal pool contains both screened surviving incumbents and unscreened new drivers; the solid $\gamma = 0$ profile is the nested no-entry boundary. Zero local gain can reflect an inactive rescue mass. All calculations are model-generated and nonempirical.



Continuity, endpoint vanishing, and eventual positivity in (50) localize a global maximizer to a compact subinterval. No uniqueness of m^* is claimed.

9.1. Theoretical numerical illustration

Figure 2 and Table 3 illustrate the uniform model at $\alpha = 1$ and $\beta = \delta = 0.8$. These values are chosen to display the mechanism and are not empirical estimates. Each thickness point re-solves the equilibrium cutoff and then optimizes p_2 conditional on p_1 before optimizing p_1 .

10. A public-supply benchmark

The Poisson model makes realized local supply latent. To separate that information friction from assignment competition, fix instead a deterministic incumbent count $n \geq 1$ observed by every

Table 3 Selected-thickness theoretical solutions.

m	p_1^*	p_2^*	a^*	M_R^*	M_F^*	$100V(m)$
1	0.3919	0.5464	0.3203	0.2132	0.1993	1.3883
5	0.2369	0.4131	0.1978	0.5978	0.5438	5.4003
10	0.1559	0.3103	0.1329	0.7552	0.6936	6.1552
20	0.0925	0.2081	0.0804	0.8624	0.8080	5.4437

Parameters are $\alpha = 1$, $\beta = \delta = 0.8$, and $\gamma = 0$. a^* is the equilibrium first-period cutoff and $100V(m)$ is the completion gain in percentage points. The table is a theoretical numerical illustration, not an empirical calibration.

agent. We evaluate one announced menu conditional on that fixed n ; the exercise removes rival-count uncertainty without introducing state-contingent pricing. Keep the timing, uniform private costs, survival, and no-entry assumption. For a conjectured cutoff a , a focal acceptor faces $X \sim \text{Bin}(n-1, a)$ accepting rivals, so her assignment share is

$$s_n(a) := \mathbb{E} \left[\frac{1}{1+X} \right] = \frac{1 - (1-a)^n}{na}, \quad s_n(0) = 1. \quad (57)$$

Universal rejection occurs against all of her rivals with probability $(1-a)^{n-1}$. Conditional on that event, each rival has cost uniform on $(a, 1)$. If terminal payment p_j is selected, a rival is both surviving and eligible with probability

$$\theta_j(a) = \frac{\alpha(p_j - a)}{1 - a};$$

hence a terminal acceptor's assignment share is $s_n(\theta_j)$ and terminal coverage is

$$C_{j,n}(a) = 1 - [1 - \theta_j(a)]^n = n\theta_j(a)s_n(\theta_j(a)). \quad (58)$$

Let $\eta_j(a)$ be the posting-conditional mass of riders who select terminal payment p_j after failure. At a positive interior cutoff, indifference of the cutoff driver is

$$s_n(a)(p_1 - a) = \delta(1-a)^{n-1}\alpha \sum_j \eta_j(a)s_n(\theta_j(a))(p_j - a). \quad (59)$$

Substitution of (58)–(59) into unconditional completion gives the exact finite-population counterpart of (20):

$$M_n(p_1, p_2; a) = (1 - p_1)n s_n(a) \left[a + \frac{p_1 - a}{\delta} \right]. \quad (60)$$

PROPOSITION 3 (Public n and local announced rescue). *Fix $n \geq 1$, $0 < \alpha \leq 1$, $0 < \delta \leq 1$, and $p \in (0, \beta)$. For every sufficiently small $\varepsilon > 0$, the menu $(p, p + \varepsilon)$ has a unique symmetric cutoff, and it satisfies*

$$a_\varepsilon = p - \kappa_n \varepsilon + o(\varepsilon), \quad \kappa_n = \frac{\delta(1-p)^{n-1}\alpha\rho(p)}{s_n(p) - \delta(1-p)^{n-1}\alpha\rho(p)} > 0. \quad (61)$$

Its right derivative of unconditional completion is

$$\left. \frac{dM_n}{d\varepsilon} \right|_{0+} = (1-p)n\kappa_n \left\{ \frac{1-\delta}{\delta} s_n(p) - p s'_n(p) \right\}. \quad (62)$$

This derivative is strictly positive for every $n \geq 1$ when $\delta < 1$. At $\delta = 1$, it is strictly positive for $n \geq 2$ and exactly zero for $n = 1$.

The denominator in (61) is positive because $s_n(p) \geq (1-p)^{n-1}$ and $\delta\alpha\rho(p) < 1$. Moreover

$$s_n(a) = \frac{1}{n} \sum_{r=0}^{n-1} (1-a)^r,$$

which is strictly decreasing for $n \geq 2$ and constant for $n = 1$. Proposition 3 rules out the claim that an unobserved realized count is necessary for the rescue gain. The first term in (62) is pure intertemporal compensation and survives at $n = 1$; the second is assignment competition and is positive exactly when $n \geq 2$. Latent supply changes posterior inference and the global policy problem, but neither local wedge. If $\alpha = 0$, then $\kappa_n = 0$ and the derivative in (62) is zero for every n .

11. Planned Empirical Calibration and Counterfactual Design

Current status. This section is a pre-specified empirical placeholder. The repository does not contain request-level observations or a protocol verifying that p_2 was announced before first-period decisions. No estimate, standard error, model fit, or empirical counterfactual is claimed in the present version. The theoretical numerical illustration in Section 9 is not an empirical calibration.

11.1. Parameter status and identifying variation

Before estimation, every model object will receive exactly one status: measured, estimated, calibrated, or fixed for sensitivity. Table 4 records the identifying variation required for the main primitives.

The initial implementation maintains the model restriction that incumbent and entrant costs share the same primitive distribution F . A later robustness extension may allow $F_I \neq F_E$, but doing so requires rederiving and recomputing the equilibrium rather than relabeling the current moments. Even under a common primitive distribution, incumbents are selected into the continuation pool by first-period rejection whereas entrants are unscreened.

11.2. Calibration criterion and equilibrium computation

Let θ collect the empirical parameters and let $\hat{g}$ contain request-level moments. The planned minimum-distance estimator is

$$\hat{\theta} \in \arg \min_{\theta \in \Theta} [\hat{g} - g(\theta)]^\top W [\hat{g} - g(\theta)]. \quad (63)$$

Table 4 Pre-specified parameter and identification ledger (design only; no estimates).

Object	Planned status	Required variation or moment	Current entry
m	measured or estimated state	Initial eligible exposure and pre-decision supply signal	DATA REQUIRED
α	measured or estimated	Fraction of rejected incumbents still eligible in period 2	DATA REQUIRED
γ	measured or estimated	New period-2 eligible drivers relative to initial expected supply	DATA REQUIRED
F	calibrated	Acceptance by exogenous payment among exposed drivers	DATA REQUIRED
G, β	estimated or bounded	Posting and continuation across price, delay, and pre-dicted coverage	DATA REQUIRED
δ	estimated or bounded	First-period acceptance response to announced p_2 holding p_1 fixed	DATA REQUIRED

A static price experiment can discipline contemporaneous acceptance but does not identify δ or the announcement channel without ex ante variation in p_2 . Persistent driver identifiers are required to separate α from γ .

Target moments include posting, first-period acceptance, universal rejection, abandonment, repetition, rescue activation, incumbent survival, fresh entry, terminal acceptance, and unconditional completion. Each evaluation of $g(\theta)$ must resolve the cutoff-WPBE for every relevant menu and market state. All roots and boundary equilibria are enumerated, and the maintained selection rule is applied before moments are aggregated. The platform policy is optimized only after this equilibrium step.

The estimation report will list parameter bounds, normalizations, starting points, global and local search methods, numerical tolerances, convergence criteria, cutoff residuals, and the weighting matrix. Parameters not point identified by the available variation will be reported as profiles, bounds, or sensitivity regions rather than arbitrary point estimates.

11.3. Fit and validation

Targeted moments and validation moments will be separated before estimation. When the data permit, a market, date, geography, or randomized arm will be held out. The principal benchmark models are flat pricing, myopic acceptance, no anticipation of p_2 , a model that pools incumbents and entrants, and alternative F_I, F_E, G specifications.

11.4. Policy counterfactuals

If the data gates are satisfied, a future empirical implementation will compare the observed policy, optimized flat pricing, optimized announced rescue, and a no-anticipation benchmark. The main outcomes are $p_1^*(m)$, $p_2^*(m)$, $M_F^*(m)$, $M_R^*(m)$, $V(m)$, first-period completion, rescue use, and delay. The extension varies α , γ , β , and δ separately so that the effects of screened surviving incumbents and unscreened entrants are not summarized by a single continuation-supply number.

Sampling and parameter uncertainty will be propagated through the entire equilibrium and policy-optimization procedure. The resampling unit will match the assignment and dependence

Table 5 Model fit and validation placeholders (no estimates).

Moment	Data	Model	Use
First-period acceptance	DATA REQUIRED	DATA REQUIRED	Target/holdout
Universal rejection	DATA REQUIRED	DATA REQUIRED	Target/holdout
Abandon/repeat/rescue shares	DATA REQUIRED	DATA REQUIRED	Target/holdout
Incumbent survival and fresh entry	DATA REQUIRED	DATA REQUIRED	Target/holdout
Terminal acceptance	DATA REQUIRED	DATA REQUIRED	Target/holdout
Unconditional completion	DATA REQUIRED	DATA REQUIRED	Target/holdout

The final table will report sample sizes, empirical uncertainty, absolute error, and whether each moment was targeted or reserved for validation.

Table 6 Counterfactual-policy placeholders (no estimates).

Policy	p_1	p_2	Completion	Rescue use
Observed/current	DATA REQUIRED	DATA REQUIRED	DATA REQUIRED	DATA REQUIRED
Optimized flat	DATA REQUIRED	same as p_1	DATA REQUIRED	0
Optimized announced rescue	DATA REQUIRED	DATA REQUIRED	DATA REQUIRED	DATA REQUIRED
No-anticipation benchmark	DATA REQUIRED	DATA REQUIRED	DATA REQUIRED	DATA REQUIRED

Completion changes will be reported in percentage points and relative to the appropriate baseline. Global optimization with $\gamma > 0$ is numerical evidence, not an extension of the no-entry global theorem.

structure, and parameters will be re-estimated inside every bootstrap draw. Confidence bands for $p_1^*(m)$, $p_2^*(m)$, and $V(m)$ will distinguish the support of observed thickness from model extrapolation.

12. Managerial interpretation and institutional scope

The fixed- p_1 solution gives a concrete menu-design rule rather than a generic instruction to raise payment after failure. A sufficiently low first payment places the request in a reject-all regime and uses the unique rescue-envelope maximizer. At an intermediate p_1 , the platform chooses a tangent rescue payment that implements the smallest feasible cutoff. Once p_1 reaches rider patience β , a separate rescue payment no longer changes positive-mass continuation behavior.

Three state variables matter operationally. Rider patience determines whether delayed completion remains valuable enough to finance rescue. Market thickness determines how much room the rescue menu has to improve on an already optimized flat payment: $V(m)$ is small when opportunities are scarce and again when flat pricing already completes almost every request. Supply composition matters even at a fixed total expected second-window count. Surviving incumbents have been screened by first-period rejection and discount waiting, whereas fresh entrants are unscreened and respond contemporaneously. Pooling the two groups into a single continuation-supply number therefore removes the very history that drives their different acceptance behavior.

The public-supply benchmark clarifies two channels. Incumbent impatience can make rescue locally productive even with one known driver; with multiple drivers, assignment competition adds a further incentive to accept early. In very thick markets the optimized rescue policy reduces failure from order $\log m/m$ under flat pricing to order $\log \log m/m$. The absolute completion difference still vanishes, so a large proportional reduction in a small residual failure rate is not a large percentage-point gain.

These implications have a defined institutional scope. The maintained transfer is paid by the rider to the selected driver and the objective is completion. Platform-funded bonuses, profit, budgets, and welfare require additional resource-cost primitives. Simultaneous acceptance with uniform selection represents a broadcast response window, not a first-response race. The equilibrium analysis is restricted to anonymous symmetric pure-cutoff WPBE. Fresh entry is included in the equilibrium and local-rescue analysis, whereas the global menu and thickness theorems impose $\gamma = 0$; no unique maximizing thickness is asserted.

13. Conclusion

An announced rescue menu changes current acceptance as well as continuation coverage. A driver who rejects p_1 may survive to receive p_2 , but waiting also exposes the driver to assignment competition. Universal rejection is therefore both the operational trigger for rescue and a signal about the realized incumbent pool and its costs. Private rider continuation makes the platform solve these supply and demand responses jointly.

In the uniform no-entry baseline and the maintained cutoff-WPBE class, every menu has a unique numeric cutoff. The conditional second-payment problem is a concave envelope, and the global menu problem reduces to a continuous one-dimensional maximum. The optimized rescue menu beats the optimized flat payment above an endogenous rider-patience threshold; with positive incumbent survival, $\beta \geq 1/2$ is sufficient at every finite m . The completion gain $V(m)$ is evaluated at the same thickness under both optimized policies. It vanishes in very thin and very thick markets and reaches a finite positive maximum, without implying a unique peak.

The survival-and-entry extension keeps screening history explicit. Drivers who rejected initially form a truncated incumbent pool and survive with probability α , whereas fresh entrants arrive at intensity γm without that first-period screen. Local rescue remains productive when its activity condition holds. If at least one optimized flat price is active, a small announced rescue strictly dominates the optimized flat policy even when both sources coexist. This condition is substantive: when it fails strictly, the same small perturbation is exactly unused. Holding either the raw continuation headcount or the price-marginal capacity fixed, a larger incumbent share strictly amplifies first-order strategic waiting within the active local-rescue region. Its marginal completion effect is

not monotone in composition; a closed-form endpoint threshold shows how the ranking can reverse with δ . The fully global menu theorem intentionally remains a no-entry result.

The empirical section specifies the data and identification needed to move from theory to calibration. In particular, credible ex ante disclosure of p_2 , request–driver exposure histories, persistent driver identifiers, and rider continuation events are required to distinguish anticipation, incumbent survival, and fresh entry. The present repository contains the measurement contract and synthetic validation tests but no observed request-level data. Consequently it reports no parameter estimate, model fit, or empirical counterfactual. Calibrated policy conclusions are left for the version in which these data gates are satisfied.

References

- Afèche P, Liu Z, Maglaras C (2023) Ride-hailing networks with strategic drivers. *Manufacturing & Service Operations Management* 25(5):1890–1908, URL <http://dx.doi.org/10.1287/msom.2023.1221>.
- Ausseil R, Pazour J, Ulmer M (2022) Supplier menus for dynamic matching in peer-to-peer transportation platforms. *Transportation Science* 56(5):1304–1326, URL <http://dx.doi.org/10.1287/trsc.2022.1133>.
- Aviv Y, Pazgal A (2008) Optimal pricing of seasonal products in the presence of forward-looking consumers. *Manufacturing & Service Operations Management* 10(3):339–359, URL <http://dx.doi.org/10.1287/msom.1070.0183>.
- Bai J, Heese HS, Tripathy M (2023) Hiding in plain sight: Surge pricing and strategic providers. *Production and Operations Management* 32(12):3837–3855, URL <http://dx.doi.org/10.1111/poms.14064>.
- Buchanan J, Gjerstad S, Porter D (2016) Information effects in uniform price multi-unit Dutch auctions. *Southern Economic Journal* 83(1):126–145, URL <http://dx.doi.org/10.1002/soej.12145>.
- Carare O, Rothkopf M (2005) Slow Dutch auctions. *Management Science* 51(3):365–373, URL <http://dx.doi.org/10.1287/mnsc.1040.0328>.
- Chen CH (2012) Name your own price at Priceline.com: Strategic bidding and lockout periods. *Review of Economic Studies* 79(4):1341–1369, URL <http://dx.doi.org/10.1093/restud/rds005>.
- Chen Y, Hu M (2020) Pricing and matching with forward-looking buyers and sellers. *Manufacturing & Service Operations Management* 22(4):717–734, URL <http://dx.doi.org/10.1287/msom.2018.0769>.
- Correa J, Montoya R, Thraves C (2016) Contingent preannounced pricing policies with strategic consumers. *Operations Research* 64(1):251–272, URL <http://dx.doi.org/10.1287/opre.2015.1452>.
- Cr  mer J, Spiegel Y, Zheng CZ (2007) Optimal search auctions. *Journal of Economic Theory* 134(1):226–248, URL <http://dx.doi.org/10.1016/j.jet.2006.03.003>.
- DoorDash (2026) How DoorDash is preparing for winter storm Fern. DoorDash News, january 23, 2026. [Online article](#).

- Feng X, Wang M (2023) Strategic driver's acceptance-or-rejection behavior and cognitive hierarchy in on-demand platforms. *Transportation Research Part E: Logistics and Transportation Review* 176:103175, URL <http://dx.doi.org/10.1016/j.tre.2023.103175>.
- Feng Y, Niazadeh R, Saberi A (2023) Two-stage stochastic matching and pricing with applications to ride hailing. *Operations Research* 72(4):1574–1594, URL <http://dx.doi.org/10.1287/opre.2022.2398>.
- Garg N, Nazerzadeh H (2022) Driver surge pricing. *Management Science* 68(5):3219–3235, URL <http://dx.doi.org/10.1287/mnsc.2021.4058>.
- Guda H, Subramanian U (2019) Your Uber is arriving: Managing on-demand workers through surge pricing, forecast communication, and worker incentives. *Management Science* 65(5):1995–2014, URL <http://dx.doi.org/10.1287/mnsc.2018.3050>.
- Horner H, Pazour J, Mitchell J (2021) Optimizing driver menus under stochastic selection behavior for ridesharing and crowdsourced delivery. *Transportation Research Part E: Logistics and Transportation Review* 153:102419, URL <http://dx.doi.org/10.1016/j.tre.2021.102419>.
- Hu B, Hu M, Zhu H (2022) Surge pricing and two-sided temporal responses in ride hailing. *Manufacturing & Service Operations Management* 24(1):91–109, URL <http://dx.doi.org/10.1287/msom.2020.0960>.
- Lee J, Li D (2023) Seller compound search for bidders. *The Journal of Industrial Economics* 71(4):1004–1037, URL <http://dx.doi.org/10.1111/joie.12355>.
- Li Z, Kuo CC (2013) Design of discrete Dutch auctions with an uncertain number of bidders. *Annals of Operations Research* 211:255–272, URL <http://dx.doi.org/10.1007/s10479-013-1331-6>.
- Loertscher S, Muir EV, Taylor PG (2022) Optimal market thickness. *Journal of Economic Theory* 200:105383, URL <http://dx.doi.org/10.1016/j.jet.2021.105383>.
- McAfee RP, McMillan J (1988) Search mechanisms. *Journal of Economic Theory* 44(1):99–123, URL [http://dx.doi.org/10.1016/0022-0531\(88\)90098-1](http://dx.doi.org/10.1016/0022-0531(88)90098-1).
- Meskar M, Aslani S, Modarres M (2023) Spatio-temporal pricing algorithm for ride-hailing platforms where drivers can decline ride requests. *Transportation Research Part C: Emerging Technologies* 153:104200, URL <http://dx.doi.org/10.1016/j.trc.2023.104200>.
- Okoth E (2024) Hard-pressed Kenyan drivers defy Uber's algorithm, set their own fares. Reuters, august 20, 2024. [Online article](#).
- Qin X, Yang H, Liu Y (2025) A two-round broadcasting matching mechanism in ride-sourcing markets: Implication and optimization. *Transportation Research Part C: Emerging Technologies* 180:105318, URL <http://dx.doi.org/10.1016/j.trc.2025.105318>.
- Reuters (2025) China's JD.com calls out anti-competitive pressures on food delivery couriers. Reuters, april 22, 2025. [Online article](#).
- Shneyerov A (2014) An optimal slow Dutch auction. *Economic Theory* 57(3):577–602, URL <http://dx.doi.org/10.1007/s00199-014-0825-z>.

- Sigg D, Hardt M, Mendler-Dünnner C (2025) Decline now: A combinatorial model for algorithmic collective action. *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*, Article 912, URL <http://dx.doi.org/10.1145/3706598.3713966>.
- Sun H, Teunter R, Hua G, Wu T (2020) Taxi-hailing platforms: Inform or assign drivers? *Transportation Research Part B: Methodological* 142:197–212, URL <http://dx.doi.org/10.1016/j.trb.2020.10.001>.
- Wu Z, Wang L, Huang F, Zhou L, Song Y, Ye C, Nie P, Ren H, Hao J, He R, Sun Z (2022) A framework for multi-stage bonus allocation in meal delivery platform. *Proceedings of the 28th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, 4195–4203, URL <http://dx.doi.org/10.1145/3534678.3539202>, arXiv:2202.10695, <https://arxiv.org/abs/2202.10695>.
- Zhang X, Miao W, Chu J, Png I (2026) The design of centralized matching systems on two-sided platforms: Evidence from the ride-hailing market. *Marketing Science* 45(4):816–843, URL <http://dx.doi.org/10.1287/mksc.2023.0561>.
- Zhao Y, Papier F, Teo CP (2024) Market thickness in online food delivery platforms: The impact of food processing times. *Manufacturing & Service Operations Management* 26(3):853–872, URL <http://dx.doi.org/10.1287/msom.2021.0354>.

Proofs and Additional Material

EC.1. Proofs for the equilibrium characterization

Proof of Lemma 1. Poisson marking splits the incumbent process into costs below and above a , with independent counts of means ma and $m(1-a)$. Thus first-period coverage is $1 - e^{-ma}$. Under Palm conditioning a focal driver's accepting rivals form a Poisson random variable Z of mean ma , and

$$\mathbb{E}[(1+Z)^{-1}] = \int_0^1 \mathbb{E}[t^Z] dt = \int_0^1 e^{-ma(1-t)} dt = \phi(ma).$$

If the focal driver waits, continuation is reached only if every rival has cost above a , an event of probability e^{-ma} . Conditional on no point below a , the above- a Poisson process is unchanged. Independent survival thins its eligible interval $(a, p_j]$ to intensity $m\alpha(p_j - a)$, while fresh eligible drivers contribute $m\gamma p_j$. Superposition gives λ_j in (1), terminal coverage C_j , and the same integral identity gives terminal assignment share $\phi(\lambda_j)$. $\square$

Proof of Proposition 1. Given a posted request, failure is independent of v , so the rider retains the posterior $U[p_1, 1]$. Comparing the two affine continuation payoffs gives v^M and the masses in (2); the stated priority rule assigns the zero-payoff and positive repeat-rescue ties. Before posting, a type $v < p_1$ has strictly negative payoff from immediate completion and cannot obtain positive delayed surplus because $\beta v - p_j < v - p_1 < 0$. A type $v \geq p_1$ obtains a nonnegative immediate payoff whenever a driver accepts and otherwise can abandon. The posting rule therefore is exactly $v \geq p_1$.

For $c \leq p_1$, equations (4)–(5) give the affine margin (6). Since $\eta^R + \eta = \rho(p_1) < 1$ whenever it is positive,

$$B(a) \geq \phi(ma) - \delta e^{-ma} \geq \phi(ma) - e^{-ma} > 0 \quad (a > 0),$$

and at $a = 0$, $B(0) \geq 1 - \delta\alpha\rho(p_1) > 0$. Thus $A - W$ is strictly decreasing in cost; types above p_1 never prefer the negative current surplus to waiting. A symmetric best response is consequently a genuine cutoff, and evaluation at its cutoff type gives (7).

At $a = p_1$, $f_m(p_1) \leq 0$. If $f_m(0) \leq 0$, the lower boundary is an equilibrium. If $f_m(0) > 0$ and $f_m(p_1) < 0$, continuity and the intermediate value theorem give an interior zero; if $f_m(p_1) = 0$, the upper boundary is an equilibrium. The aggregate continuation payoff is continuous across a rider switch because tied actions have equal coverage-weighted utility; when all continuation coverage is zero the driver's wait payoff is zero. Hence f_m is continuous. The same argument shows that any convergent sequence of eligible boundary points or zeros has an eligible limit, so the nonempty equilibrium set is closed in $[0, p_1]$ and therefore compact. The separately stated $p_1 = 0$ and $(1, 1)$ conventions close the remaining endpoints. Formula (9) is continuous on the compact equilibrium set, making the minimum in (10) well defined. $\square$

Proof of Proposition 2. For $a < p$, the cutoff margin is $(p - a)$ times the bracket displayed below (13). If $a > 0$, then $\phi(ma) > e^{-ma}$ and $\delta\alpha\rho(p)\phi(m\{\alpha(p - a) + \gamma p\}) \leq 1$, so the bracket is strictly positive. At $a = 0$, strictness follows from $\rho(p) < 1$. Hence no cutoff below p is sustainable, while $a = p$ makes the cutoff type indifferent and is the unique numeric cutoff. At that cutoff no incumbent with cost below p remains. Terminal completion can only come from fresh drivers of intensity γmp , and the mass of posting riders with nonnegative delayed surplus is $[1 - p/\beta]^+/(1 - p)$. Adding initial and terminal completion yields (12). The continuous objective on $[0, 1]$ attains the flat maximum.

□

Proof of Lemma 2. For $p < \beta$,

$$Q_\gamma^F(m, p) = q_0(m, p) + e^{-mp}(1 - p/\beta)(1 - e^{-\gamma mp}), \quad (\text{EC.1})$$

whereas $Q_\gamma^F(m, p) = q_0(m, p)$ for $p \geq \beta$. The one-sided derivatives at the rider-continuation kink satisfy

$$Q_{\gamma,+}^{F'}(m, \beta) - Q_{\gamma,-}^{F'}(m, \beta) = \frac{e^{-m\beta}(1 - e^{-\gamma m\beta})}{\beta} > 0. \quad (\text{EC.2})$$

A local maximum requires a nonnegative left derivative and a nonpositive right derivative, so this upward jump excludes $p = \beta$. Moreover, $Q_{\gamma,+}^{F'}(m, 0) = m(1 + \gamma) > 0$, and the endpoint $p = 1$ has zero completion, so neither endpoint is optimal.

Direct differentiation gives

$$q_{0,pp}(m, p) = -me^{-mp}\{2 + m(1 - p)\} < 0. \quad (\text{EC.3})$$

Thus q_0 is strictly concave, and its first-order condition is $e^{mp} = 1 + m(1 - p)$. Solving it gives $p^0(m)$ in (F1). The left side minus the right side is negative at zero and strictly positive at $p = 1/2$ because $e^{m/2} > 1 + m/2$; hence $0 < p^0(m) < 1/2$. On $[\beta, 1]$ the only possible global maximizer is therefore $p^0(m)$, and it lies in that interval only if $\beta < p^0(m)$. This proves (F2).

It remains to verify the stated nonuniqueness possibility. Fix $\beta < p^0(m)$ and extend the objective continuously to $\gamma = 0$. Let

$$B(\gamma) = \max_{p \in [0, \beta]} Q_\gamma^F(m, p).$$

The maximum theorem makes B continuous. At zero, $B(0) = q_0(m, \beta) < q_0(m, p^0)$. For $p_\gamma = (\log \gamma)/(\gamma m)$, which lies below β for large γ , equation (EC.1) gives $Q_\gamma^F(m, p_\gamma) \rightarrow 1 > q_0(m, p^0)$. The intermediate value theorem therefore gives some $\gamma^\dagger > 0$ with $B(\gamma^\dagger) = q_0(m, p^0)$. At this value p^0 and a point strictly below β are both global flat maximizers; the latter cannot equal β because $q_0(m, \beta) < q_0(m, p^0)$. □

Proof of Theorem 1. Put $d = p - a$. Under $(p, p + \varepsilon)$ the two terminal intensities are

$$\lambda_1 = m(\alpha d + \gamma p), \quad \lambda_2 = \lambda_1 + m(\alpha + \gamma)\varepsilon. \quad (\text{A1})$$

If $\eta_\varepsilon(a)$ is the rescue mass, interior cutoff indifference is

$$\begin{aligned} \phi(ma)d &= \delta\alpha e^{-ma} \{[\rho - \eta_\varepsilon]\phi(\lambda_1)d \\ &\quad + \eta_\varepsilon\phi(\lambda_2)(d + \varepsilon)\}. \end{aligned} \quad (\text{A2})$$

Rearrange and use $\phi(\lambda_j) \leq 1$, $\eta_\varepsilon \leq \rho$, and $\phi(ma) \geq e^{-ma}$. The coefficient on d is at least $e^{-ma}(1 - \delta\alpha\rho) > 0$, which gives (L4). The same weak boundary inequality excludes a reject-all cutoff for small ε ; the upper cutoff already obeys the bound. Thus every equilibrium is localized, not only a selected branch.

On the compact rescaling $y = d/\varepsilon$, (A1) gives

$$\frac{C_2 - C_1}{\varepsilon} \rightarrow mE(\alpha + \gamma), \quad C_2 \rightarrow 1 - E.$$

The rider switch therefore converges uniformly to $\bar{v}$ in (L2), and the rescue mass converges to $\eta^0 > 0$. Divide (A2) by ε and pass to the limit:

$$[\sigma - \delta R\alpha\rho\ell]y - \delta R\alpha\eta^0\ell = 0. \quad (\text{A3})$$

Its slope is at least $\sigma - R > 0$. The rescaled margin converges in C^1 to this affine function, so it has a unique zero for all small ε ; (A3) yields (L3).

It remains to establish the strict sign in (L6). First set $\delta = 1$ and call the resulting expression T_1 . The case $\gamma = 0$ gives $T_1 = \alpha(\sigma - R) > 0$. For $\gamma > 0$, set

$$u = e^x - 1, \quad t = 1 - E, \quad A = \alpha + \gamma, \quad \Delta = 1 - E(1 + \gamma), \quad B = 1 - \rho\Delta.$$

Since $\sigma = Ru/x$ and $\ell = t/(\gamma x)$,

$$T_1 = \frac{R}{\gamma x}Q, \quad Q = E\gamma Au - \alpha tB. \quad (\text{A4})$$

If $\Delta \leq 0$, then $B \leq E(1 + \gamma)$, $\gamma u > t$, and $A \geq \alpha(1 + \gamma)$, so $Q > 0$. If $\Delta > 0$ and $\gamma \leq 1$, convexity gives $t \leq E\gamma u$ and $B \leq 1$, whence $Q \geq \alpha(E\gamma u - t) + E\gamma^2 u > 0$.

Finally suppose $\Delta > 0$ and $\gamma > 1$. Activity implies the weaker strict bound

$$\rho > \frac{t}{t + xEA}.$$

It implies

$$Q \geq \frac{E}{t + xEA}I(\alpha), \quad I(\alpha) = \gamma Au(t + xEA) - \alpha t\{xA + t(1 + \gamma)\}. \quad (\text{A5})$$

The coefficient on α^2 in I is $x(\gamma Eu - t) < 0$, because $e^{\gamma x} - 1 > \gamma(e^x - 1)$. Hence the minimum of this concave quadratic on $[0, 1]$ is at an endpoint. Clearly $I(0) > 0$. At $\alpha = 1$, write $H = t + xE(1 + \gamma)$ and $d = \gamma u - t$. Then

$$\frac{I(1)}{1 + \gamma} = dH - tx\Delta. \quad (\text{A6})$$

Let $w = \gamma x$. Since $\Delta > 0$, $w > \log(1 + \gamma)$; moreover

$$d > \frac{\gamma x^2}{2} = \frac{xw}{2}, \quad \Delta = 1 - (1 + \gamma)e^{-w} < \frac{w}{2}.$$

The last inequality follows by minimizing $w/2 - 1 + (1 + \gamma)e^{-w}$ on $[\log(1 + \gamma), \infty)$. Thus $d > x\Delta$ and, since $H > t$, (A6) is positive. This proves $Q > 0$ in all cases and hence $T_1 > 0$. For general $0 < \delta \leq 1$,

$$T_\delta = T_1 + (1 - \delta)R\alpha\ell\{1 - \rho + \rho E(1 + \gamma)\} > 0.$$

Along the cutoff branch, write continuation coverage as $S_\varepsilon = \rho C_1 + \eta_\varepsilon(C_2 - C_1)$. Direct differentiation at zero gives

$$S'_0 = mE\{\rho\alpha\kappa + \eta^0(\alpha + \gamma)\}.$$

Differentiating (9), substituting (L3), and collecting terms yields exactly (L5)–(L6). All remaining factors are positive, completing the proof. $\square$

Proof of Theorem 2. Condition (L9) is exactly the activity condition $\bar{v} < 1$ in Theorem 1, evaluated at $p = p^*$. That theorem gives a unique cutoff and the expansion

$$M_m^{\alpha, \gamma}(p^*, p^* + \varepsilon; a_\varepsilon) = M_m^{\alpha, \gamma}(p^*, p^*; p^*) + \Lambda(p^*)\varepsilon + o(\varepsilon), \quad (\text{EC.4})$$

where $\Lambda(p^*)$ is the strictly positive expression in (L5). Since $p^* \in \mathcal{P}_\gamma^F(m)$, the first term on the right is $F_\gamma^*(m)$. Uniqueness makes the conservative minimum in (10) equal to this branch value. A sufficiently small positive ε therefore strictly improves on $F_\gamma^*(m)$, and the supremum in (11) is at least that strictly higher value. This proves (L10)–(L11) without requiring the supremum to be attained.

Now suppose $\mathcal{A}_{\alpha, \gamma}(p) < 0$. The localization argument leading to (L4) uses only $\eta_\varepsilon \leq \rho$ and hence still applies. On its compact rescaling $0 \leq (p - a)/\varepsilon \leq \delta\alpha\rho/(1 - \delta\alpha\rho)$, the repeat-rescue switch converges uniformly to $\bar{v} > 1$. Thus $\eta_\varepsilon = 0$ throughout that set for all small ε . For every $a < p$, cutoff indifference then has the strict flat-sign factor

$$(p - a) [\phi(ma) - \delta\alpha e^{-ma}\rho(p)\phi(m\{\alpha(p - a) + \gamma p\})] > 0. \quad (\text{EC.5})$$

Consequently no cutoff below p is sustainable, while $a = p$ is eligible and unique. Rescue is unused and the exact completion is $Q_\gamma^F(m, p)$. At equality the limiting rescue mass is zero and this first-order argument does not determine the sign. $\square$

Proof of Theorem 3. Write $x = mp$, $R = e^{-x}$, and $\sigma = \phi(x)$. The local expansion (L3) can be written for any active composition as

$$\kappa = \frac{\delta R q \eta}{\sigma - \delta R \rho(p) q}, \quad q = \alpha \phi(\gamma x). \quad (\text{EC.6})$$

The denominator is strictly positive because $q \leq \alpha \leq 1$, $\rho(p) < 1$, and $\sigma > R$.

Along the fixed-headcount path, $\gamma'_h(\alpha) = -(1-p)$ and $A'_h(\alpha) = p$. Therefore

$$\frac{d}{d\alpha} \frac{e^{\gamma_h(\alpha)x} - 1}{A_h(\alpha)} = \frac{-(1-p)x e^{\gamma_h(\alpha)x} A_h(\alpha) - p\{e^{\gamma_h(\alpha)x} - 1\}}{A_h(\alpha)^2} < 0. \quad (\text{EC.7})$$

It follows that η_h is strictly increasing. Moreover,

$$\phi'(z) = \frac{e^{-z}(1+z) - 1}{z^2} < 0 \quad (z > 0), \quad (\text{EC.8})$$

so the decrease in γ_h makes ℓ_h increase and makes $q_h(\alpha) = \alpha \ell_h(\alpha)$ strictly increase. For positive q and η , the right side of (EC.6) satisfies

$$\frac{\partial \kappa}{\partial q} = \frac{\delta R \eta \sigma}{\{\sigma - \delta R \rho(p) q\}^2} > 0, \quad \frac{\partial \kappa}{\partial \eta} = \frac{\delta R q}{\sigma - \delta R \rho(p) q} > 0. \quad (\text{EC.9})$$

This proves part (i), including comparisons from $\alpha = 0$ by continuity.

Along the fixed-capacity path, $\gamma'_s(\alpha) = -1$ while $\alpha + \gamma_s(\alpha) = s$. Hence

$$\frac{d}{d\alpha} \frac{e^{\gamma_s(\alpha)x} - 1}{s} = -\frac{x e^{\gamma_s(\alpha)x}}{s} < 0. \quad (\text{EC.10})$$

Thus η_s , ℓ_s , and $\alpha \ell_s$ all increase strictly on the active set. Equation (EC.9) proves part (ii). $\square$

Proof of Theorem 4. Condition (L20) is $\eta_s(0) > 0$. Since $\eta_s(\alpha)$ increases in α , the full path is active. At the all-entrant endpoint $(\alpha, \gamma) = (0, s)$, cutoff erosion is zero and substitution into (L5)–(L6) gives the first expression in (L22). At the all-incumbent endpoint $(\alpha, \gamma) = (s, 0)$, we have $E = \ell = 1$ and $\eta^0 = \rho$, so the same formula gives the second expression in (L22).

Positive entry raises the repeat-rescue switch strictly above p/β . Thus (L20) implies $0 < \eta_E < \rho$, and $0 < H < 1$. Also $s\rho < 1$, so every denominator below is positive. Removing the common positive factor $m(1-p)Rs\rho$ from $L_s(s) - L_s(0)$ leaves the sign of

$$\frac{\sigma - \delta R}{\sigma - \delta Rs\rho} - H = \frac{\sigma(1-H) - \delta R(1-Hs\rho)}{\sigma - \delta Rs\rho}. \quad (\text{EC.11})$$

The numerator crosses zero exactly at (L23), proving (L24). This comparison ranks only the two pure endpoints. The values in (L25), obtained from the same exact local formula at an interior composition, establish that the path itself need not be monotone. $\square$

EC.2. Proofs for global no-entry design

Proof of Theorem 5. The cancellation leading to (16) follows by substituting the rider switch type into the sum of repeat and rescue coverage. Inequality (18) excludes the repeat branch and its kink from the zero set. Every zero is therefore a smooth active-rescue zero. With the notation preceding (19), its root equation, $B - b = p_2 - \beta p_1 > 0$, and $\delta \leq 1$ imply $u < C(z)/m$. Consequently

$$\frac{1}{u} - \frac{k}{e^{kz} - 1} > m \frac{1 - \alpha e^{-kz}}{1 - e^{-kz}} \geq m > \frac{h'(a)}{h(a)},$$

where $h'(a)/h(a) < m$ follows from $h(a) = \int_0^m e^{ta} dt$. Thus (19) is strictly negative and every zero crosses downward. Also $\mathcal{D}_{12}^\delta(p_1) = -\delta(\beta - p_2)C(p_2 - p_1)/B_1 < 0$. A continuous function whose every zero is a strict downward crossing has at most one zero. The endpoint signs give exactly the two cases stated in the theorem, including the equality case at zero. Flat and repeat-only menus use the strict flat inequality and have cutoff p_1 ; if $p_1 \geq \beta$, no positive mass continues; if $\alpha = 0$, waiting has zero payoff; and $p_1 = 0$ has only the lower numeric cutoff. These observations cover every no-entry menu. $\square$

Proof of Theorem 6. For a target cutoff a , active-rescue implementability is exactly (31). The strict flat inequality makes its right side larger than $A(a, p_1)$, so an implementer is strictly inside the rescue region. The function $A(a, p_2)$ is strictly concave because

$$A_{p_2 p_2}(a, p_2) = -e^{-k(p_2 - a)}[2k + k^2(\beta - p_2)] < 0,$$

and its unique maximizer is $Q(a)$. At $a = 0$, reject-all is implementable iff $A(0, p_2) \geq \beta(1 - p_1)mp_1/\delta$. Its largest possible left side is S_0 , so the condition is feasible exactly when $p_1 \leq p_z$; the strict flat inequality at $(a, p_1) = (0, Q_0)$ gives $0 < p_z < Q_0 < 1/2$.

If $p_1 \leq p_z$, any positive-cutoff completion is below $(1 - p_1)mp_1/\delta \leq S_0/\beta$, whereas reject-all completion is $A(0, p_2)/\beta \leq S_0/\beta$. Strict concavity makes $p_2 = Q_0$ the unique maximizer, including at $p_1 = p_z$. If $p_z < p_1 < \beta$, maximize A pointwise in p_2 and define the envelope margin

$$\overline{\mathcal{D}}_{p_1}(a) = (p_1 - a)h(a) - \delta S(a)/[\beta(1 - p_1)].$$

It is positive at zero and negative at p_1 . The same logarithmic derivative argument as in Theorem 5 shows that every envelope zero crosses downward, hence (30) has exactly one root $a^*(p_1)$. Any menu implementing a satisfies $A(a, p_2) \leq S(a)$, so its unique cutoff cannot lie below $a^*(p_1)$. Completion in (20) decreases with a positive cutoff; therefore the tangent menu $p_2 = Q(a^*)$ is the unique maximizer. For $p_1 \geq \beta$, no positive mass uses rescue and all $p_2 \geq p_1$ are outcome-equivalent to the flat payment. This proves all three regimes and attainment. $\square$

Proof of Theorem 7. Let $g_a(p_1) = (1 - p_1)(p_1 - a)$. Equation (26) implies $Q(a) < (1 + a)/2$, so g_a is strictly increasing on $(a, Q(a))$. The strict flat inequality gives $g_a(Q(a)) > R_\delta(a)$, while $g_a(a) = 0 < R_\delta(a)$; hence (32) has exactly one root in that interval, the lower branch (33). If two distinct cutoffs had the same value of P_δ , the same fixed- p_1 envelope would have two downward-crossing zeros. This also excludes $P_\delta(a) = P_\delta(0)$ for $a > 0$. Thus the continuous map P_δ is strictly increasing from $[0, \beta]$ onto $[p_z, \beta]$.

The three fixed-price regimes now give exactly (35). Both objectives are continuous on fixed compact intervals, so each maximum and the original policy maximum are attained. The flat objective is strictly concave; its first-order condition is $e^{mp_F} - 1 = m(1 - p_F)$, whose unique root satisfies $p_F < 1/2$. For any $p_1 < p_2 < \beta$, Theorem 5 gives a cutoff below p_1 ; equations (20)–(21) then make completion strictly larger than $F_m(p_1)$. If $\beta \geq 1/2$, apply this comparison at $p_F < \beta$. The reject-all boundary and $p_1 \geq \beta$ flat region do not beat F_0^* , whereas the strict comparison does, so every global maximizer is a strict interior menu parameterized as in (36). No step requires uniqueness of the outer argmax. $\square$

Proof of Theorem 8. First fix a strict menu $p_1 < p_2$. If $\beta \leq p_2$, rescue has no positive-mass use and the strict flat inequality forces cutoff p_1 , so the menu value is $F_m(p_1)$. If $p_2 < \beta_1 < \beta_2$, the coefficients $(\beta - z)/[\beta(1 - p_1)]$, $z \in \{p_1, p_2\}$, weakly increase in β and the active rescue coefficient increases strictly. At the β_1 cutoff this strictly lowers the scalar margin for β_2 . Global downward crossing therefore moves the β_2 cutoff strictly left, unless it is already zero. Equations (20)–(21) then strictly increase completion. At a zero cutoff the value is $(1 - p_2/\beta)C(p_2)$ and has positive derivative. As $\beta \downarrow p_2$, any cutoff limit below p_1 would contradict (18), so the menu value is continuous at p_2 .

Continuity of the optimized value follows from the fixed coordinate $a = \beta x$, $x \in [0, 1]$. The unique solution of $e^{ky} - 1 = k\{\beta(1 - x) - y\}$ and all objects $Q, S, P_{\beta,\delta}, J_{\beta,\delta}$ are jointly continuous, with the removable extensions described before (39). The restricted flat term is jointly continuous after writing $p = \beta + (1 - \beta)s$. The maximum theorem applied to these fixed compact sets makes $D(m; \beta)$ continuous.

Flat policies give $D \geq F_0^*$. If strict gain holds at β_1 , attainment supplies an optimal strict menu with $p_2 < \beta_1$. Reusing it at any $\beta_2 > \beta_1$ and the preceding fixed-menu result gives strict gain and a strictly larger optimized value. Thus the gain set is an open upper interval and has the form (40). An active menu has $p_2 < \beta$ and completion at most $1 - e^{-mp_2} < 1 - e^{-m\beta}$; this proves the positive lower bound in (41). If $\beta > p_F$, a strict menu at first payment p_F beats the flat optimum by (22), proving the upper bound. Finally, the second maximum in (35) never exceeds F_0^* , so strict gain is equivalent to the scalar test (42). If $\alpha = 0$, no incumbent can complete after waiting and the dynamic and flat values coincide. $\square$

EC.3. Proofs for thickness and the public-supply benchmark

Proof of Theorem 9. Equations (51)–(53) are the fixed-coordinate form of (26)–(34). The inverse of $z \mapsto z + \log(1 + z)$ is continuous. Since multiplying the old envelope term by $0 < \delta \leq 1$ only enlarges the lower-root discriminant, the strict envelope bound keeps that discriminant positive. Thus J_m^δ is jointly continuous on every compact m -interval times $[0, \beta]$. Berge’s maximum theorem applied to (35) proves part (i).

For $m \downarrow 0$, series inversion, uniformly in $a \in [0, \beta]$, gives

$$y_m(a) = \frac{\beta - a}{2} + O(m), \quad \frac{S_m(a)}{m} = \frac{\alpha(\beta - a)^2}{4} + O(m).$$

Consequently $P_m^\delta \rightarrow P_0^\delta$ and $J_m^\delta/m \rightarrow H_0^\delta$ uniformly. If $\beta > 1/2$ and $\delta < 1$, continuity and the endpoint signs give an $a \in (0, 1/2)$ with $P_0^\delta(a) = 1/2$. The value displayed after (53) is then strictly above $1/4$. Hence $G_{\alpha, \beta, \delta} > 1/4$, and uniform convergence together with $F_0^*(m)/m \rightarrow 1/4$ proves (45).

If $\delta = 1$, the limiting objective is $P_0^1(a)[1 - P_0^1(a)]$. Its unique maximum for $\beta > 1/2$ occurs at $P_0^1(a_0) = 1/2$, equivalently $\alpha(\beta - a_0)^2 = \beta(1 - 2a_0)$. Uniform second-order expansion gives

$$D_{\alpha, 1, 0}^*(m) = \frac{m}{4} - \frac{a_0}{8}m^2 + o(m^2), \quad F_0^*(m) = \frac{m}{4} - \frac{1}{16}m^2 + o(m^2),$$

which proves (46).

Now let $\beta = 1/2$. The limiting objective reaches $1/4$ only at $a = 1/2$, with a quadratic loss away from that endpoint. A uniform comparison with the endpoint therefore localizes every optimizer as $a = 1/2 - cm$ with bounded c . Expanding (51)–(53) on this compact rescaling gives (54), uniformly in c . Its strictly concave quadratic is maximized at $c^* = 1/(16A_\delta)$. Since

$$F_0^*(m) = \frac{m}{4} - \frac{m^2}{16} + \frac{11}{768}m^3 + O(m^4),$$

the difference of the cubic coefficients is

$$\frac{1}{256A_\delta} - \frac{1}{256} = \frac{\alpha(1 - \delta)}{512A_\delta},$$

proving (47). At $\delta = 1$ this difference is zero. The next uniform order is given by (54a)–(54b). The strict quadratic loss in f_2 and local Lipschitz control of f_3 imply

$$\max_c \{f_2(c) + mf_3(c) + o(m)\} = f_2(1/16) + mf_3(1/16) + o(m).$$

Since $f_3(1/16) = -3/1024 + \alpha/2048$, subtracting the flat expansion proves (48) without assuming a higher-order expansion of the optimizer itself.

If $\beta < 1/2$, $P_0^\delta(a) \geq a$ and $\alpha \leq 1$ imply the bound stated after (54). Its right side is increasing on $[0, \beta]$ and ends at $\beta(1 - \beta) < 1/4$. Uniform convergence and $p_F(m) \rightarrow 1/2 > \beta$ make the flat term the exact winner for all sufficiently small m , proving (49).

For $m \rightarrow \infty$, (55) is exact. With $x = \log \log m$ and $a = x/m$, the tangent construction gives $p_2 = O(\log m/m)$, $p_1 - a = O(\delta a e^{-x})$, and $1 - T = O(\log m/m)$, yielding the feasible upper loss $(1 + o(1)) \log \log m/m$. For the policy-uniform lower bound, if $Q_m(a) \leq \beta/2$, then

$$m[1 - J_m^\delta(a)] \geq x(1 - e^{-x}) + \frac{\log(1 + \alpha m \beta/2)}{\alpha \beta} e^{-x}.$$

Its minimum is asymptotic to $\log \log m$. If $Q_m(a) > \beta/2$, the envelope bound makes the first loss in (55) larger. This proves the dynamic rate. The flat first-order condition gives $1 - F_0^*(m) \sim \log m/m$; subtraction proves the value rate. The lower bound uses only $P_m^\delta(a) \geq a$. For the feasible upper construction, $R_\delta = \delta R_1$ implies $P_m^\delta(a) \leq P_m^1(a)$, while $T_m(a)$ is independent of δ . Hence the estimate is uniform over $\delta \in (0, 1]$ for each fixed positive (α, β) ; only sequences with $\alpha \downarrow 0$ or $\beta \downarrow 0$ fall outside this uniformity. $\square$

Proof of Corollary 1. The thin regimes imply $V(m) \rightarrow 0$ as $m \downarrow 0$, and (50) gives both $V(m) \rightarrow 0$ and $V(m) > 0$ for all sufficiently large finite m . Choose one such point. Endpoint vanishing makes the value outside some compact subinterval smaller than half its value at that point; continuity supplies a positive global maximizer on the compact interval. Equation (49) directly contradicts strict increase near zero when $\beta < 1/2$. No analogous derivative or optimizer-switch control has been proved for $\beta \geq 1/2$. $\square$

Proof of Proposition 3. The binomial identity in (57) follows by integrating $\mathbb{E}[t^X] = (1 - a + at)^{n-1}$. Given universal rejection, each rival cost is uniform on $(a, 1)$, giving θ_j , (58), and cutoff indifference (59). Conditional on posting, completion is

$$1 - (1 - a)^n + (1 - a)^n \sum_j \eta_j C_{j,n}.$$

Substitute (58) into the second term and use (59); the expression becomes $ns_n(a)[a + (p_1 - a)/\delta]$. Multiplication by posting mass $1 - p_1$ proves (60).

For an arbitrary nearby interior cutoff put $d = p - a$. Rearranging its indifference condition gives

$$\begin{aligned} [s_n(a) - \delta(1 - a)^{n-1} \alpha \{(\rho - \eta)s_n(\theta_1) + \eta s_n(\theta_2)\}]d \\ = \delta(1 - a)^{n-1} \alpha \eta s_n(\theta_2) \varepsilon. \end{aligned}$$

Because $s_n(a) \geq (1 - a)^{n-1}$, $s_n(\theta_j) \leq 1$, and $\eta \leq \rho$, every equilibrium satisfies

$$0 \leq d \leq \frac{\delta \alpha \rho}{1 - \delta \alpha \rho} \varepsilon.$$

The analogous boundary inequality excludes $a = 0$ for small ε ; at $a = p$ the cutoff type strictly prefers waiting. Thus all equilibria are interior and uniformly localize at p .

Write $a = p - \kappa\varepsilon + o(\varepsilon)$. Terminal coverage is first order in ε , the rider switch converges uniformly to p/β , and the rescue mass converges to $\rho(p)$. Divide cutoff indifference by ε and take the limit:

$$s_n(p)\kappa = \delta(1-p)^{n-1}\alpha\rho(p)(1+\kappa).$$

The denominator is positive because $s_n(p) \geq (1-p)^{n-1}$ and $\delta\alpha\rho(p) < 1$, yielding (61). Differentiating (60) along the branch yields (62). The rescaled cutoff margin converges in C^1 on the compact localization interval to the displayed affine equation, whose slope is the positive denominator in (61). Existence from the general cutoff argument and this one-crossing limit give local uniqueness. Finally, $s_n(a) = n^{-1} \sum_{r=0}^{n-1} (1-a)^r$ is strictly decreasing iff $n \geq 2$, which proves the sign statement.

□